

小腸・大腸疾患



名古屋市立大学 消化器代謝内科学
尾関 啓司

本日の内容

- 小腸内視鏡についての最近の知見
- 小腸疾患の診断と治療 各論
- 新しい小腸疾患診療の展開

小腸画像的検査方法①

➤小腸透視検査：

➤CT・MRI検査：

CT enterography(CTE)・MR enterography(MRE)

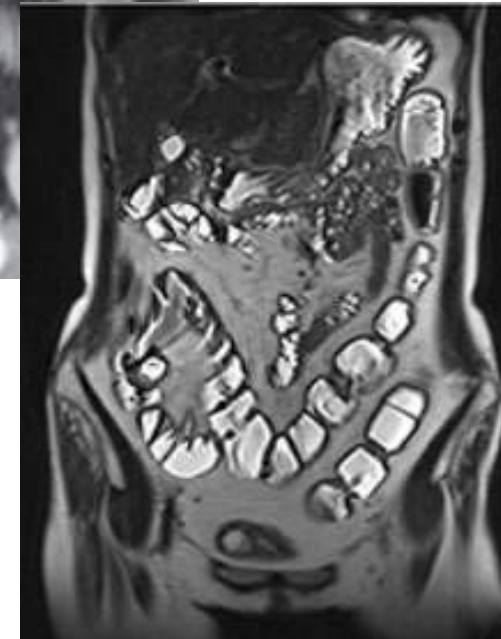
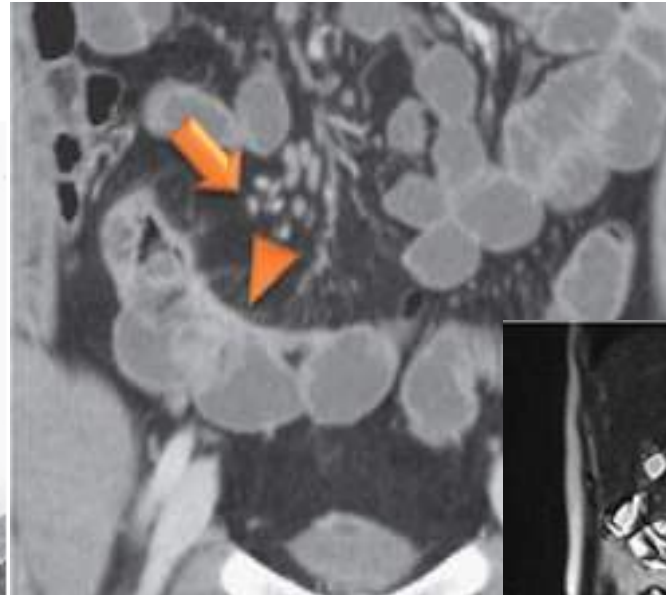
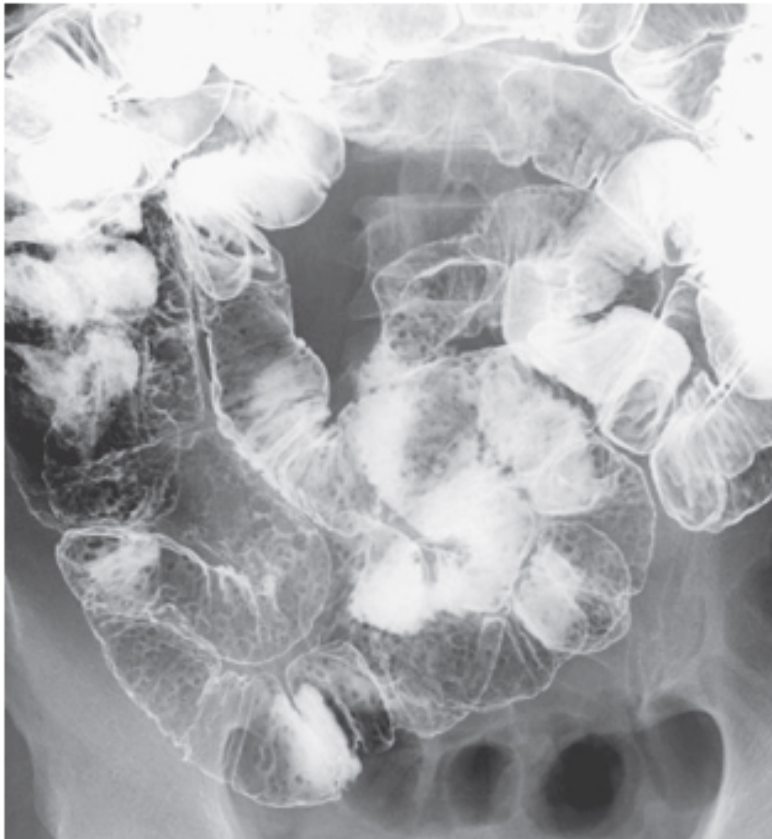
➤小腸内視鏡検査：

小腸カプセル内視鏡(CE)

デバイスアシスト内視鏡(DAE); バルーンアシスト内視鏡(DBE)
スパイラル内視鏡

➤その他

小腸画像的検査方法②



日本内科学会雑誌 第100巻 1号 特集 小腸疾患 診断と治療の進歩

日本消化器病学会雑誌 第112巻 7号 クロウン病診療におけるCTによる画像診断の実際

MR enterographyによるクローン病の評価 http://www.innervision.co.jp/ad/suite/hitachi/sup201609/topics_lecture1

小腸内視鏡ガイドライン各種①



Small-bowel capsule endoscopy and device-assisted enteroscopy for diagnosis and treatment of smallbowel disorders: European Society of Gastrointestinal Endoscopy (ESGE) Clinical Guideline Endoscopy 2015

小腸内視鏡ガイドライン各種②

Digestive Endoscopy 2017; 29: 519–546

doi: 10.1111/den.12883

Guidelines

Clinical Practice Guideline for Enteroscopy

Hironori Yamamoto,¹ Haruhiko Ogata,^{1,2} Takayuki Matsumoto,^{1,3} Naoki Ohmiya,^{1,4} Kazuo Ohtsuka,^{1,3} Kenji Watanabe,^{2,4} Tomonori Yano,^{1,4} Toshiyuki Matsui,^{1,3} Kazuhide Higuchi,^{1,2} Tetsuya Nakamura^{1,2} and Kazuma Fujimoto¹

¹Japan Gastroenterological Endoscopy Society, ²Japanese Society of Gastroenterology, ³Japanese Gastroenterological Association, and ⁴Japanese Association for Capsule Endoscopy

Clinical Practice Guideline for Enteroscopy Digestive Endoscopy 2015

ガイドライン

小腸内視鏡診療ガイドライン

山本博徳¹⁾ 緒方晴彦^{1),4)} 松本主之^{1),2)} 大宮直木^{1),3)} 大塚和朗^{1),2)}
渡辺憲治^{3),4)} 矢野智則^{1),3)} 松井敏幸^{1),2)} 樋口和秀^{1),4)} 中村哲也^{1),4)}
藤本一真¹⁾

1) 日本消化器内視鏡学会, 2) 日本消化器病学会, 3) 日本消化管学会,
4) 日本カプセル内視鏡学会

小腸内視鏡検査の適応疾患

診療部門	関連疾患		適応症	病変
消化器内科 消化器外科 光学診療部	炎症性腸疾患 消化管ポリポース 小腸腫瘍 蛋白漏出性胃腸症 吸収不良症候群		原因不明消化管出血	腫瘍性(腫瘍・ポリープ) 炎症性(潰瘍・びらん) 血管性(血管拡張等)
血液内科	鉄欠乏性貧血			
循環器内科	心筋梗塞	NSAID起因性腸炎		
神経内科	脳梗塞			
整形外科	リウマチ			
心療内科	うつ			
腎臓内科 泌尿器科	腎不全(透析)	消化管異常		
肝臓科	肝硬変・肝がん	門脈圧亢進性胃腸炎		

カプセル内視鏡

カプセル内視鏡の開発

brief communications

Wireless capsule endoscopy

The discomfort of internal gastrointestinal examination may soon be a thing of the past.

We have developed a new type of endoscopy, which for the first time allows painless endoscopic imaging of the whole of the small bowel. This procedure involves a wireless capsule endoscope and we describe here its successful testing in humans.

The invention of fibre-optic endoscopy¹ made visualization of the whole stomach, upper small bowel and colon possible. The procedures used to examine these (gastroscopy, small-bowel endoscopy and colonoscopy, respectively) cause discomfort because they require flexible, relatively wide cables to be pushed into the bowel — these cables carry light by fibre-optic bundles, power and video signals. Small-bowel endoscopy in particular is constrained by problems of discomfort and limitations of how far enteroscopes can be advanced into the small bowel. There is a clinical need for improved methods of examining the small bowel and colon, especially in patients with recurrent gastrointestinal bleeding.

The invention of the transistor made it possible to design swallowable electronic radio-telemetry capsules for the study of gastrointestinal physiological parameters. These capsules were first reported in the 1950s and were used to study the



was initially associated with CMOS image sensors and has allowed CMOS chips to achieve an image quality comparable to those of charge-coupled device image sensors², but using much less power.

Advances in ASIC design allowed the integration of a very small video transmitter of sufficient power output, efficiency and bandwidth into the capsule. Synchronous switching of the LEDs, the CMOS sensor and the ASIC transmitter minimize power consumption. By careful design of the optics, we were able to eliminate internal reflections which are a common problem when the illumination and imager are incorporated under the same dome.

With ethical committee approval, the first studies were performed on ten normal human volunteers. The capsule was easily swallowed and caused no discomfort. Propelled by peristalsis (see Supplementary Information), it successfully transmitted video images (Fig. 1) from the stomach, small bowel and caecum (mean gastric transit time was 80 min, range 17–280 min; mean small-bowel transit time was 90 min, range 45–140 min; mouth-to-evacuation time was 24 h, range 10–48 h). High-quality images were received throughout the video



1. オプティカルドーム
2. レンズホルダー
3. レンズ
4. LED
5. CMOS
6. バッテリー
7. ASIC 送信機
8. ターンコイルアンテナ

10×13mm

Wireless capsule endoscopy
Nature 2000 Iddan G et al.

カプセル内視鏡の種類

Table 1 Technical specifications of various video capsule platforms

	Manufacturer	Length (mm)	Diameter (mm)	Weight (g)	Cameras (n)	FPS	Viewing angle (deg)	Battery life (h)
PillCam SB3	Medtronic	26.2	11.4	3	1	2–6	156	8+ or 12+
PillCam Crohn's Capsule	Medtronic	32.3	11.6	2.9	2	4–35	168	10+
PillCam UGI	Medtronic	32.3	11.6	2.9	2	18–35	172	1.5
PillCam COLON 2	Medtronic	32.3	11.6	2.9	2	4–35	172	10+
EndoCapsule	Olympus	26	11	3.5	1	2	145	12+
MiroCam	Intromedic	24.5	10.8	3.25	1	3	170	11+
OMOM	Jinshan	27.9	13	6	1	2	140	6–8+

Video Capsule Endoscopy and Device-Assisted Enteroscopy
Digestive Diseases and Sciences 2022



CapsoCam
CapsoVision



各社ホームページから引用

CEの欠点；**滞留**

- 滞留の定義はカプセル内視鏡が2週間以上体内に留まる、もしくは内視鏡的、外科的に回収されなければ体外排泄が望めない状態をいう。
- OGIB（原因不明の消化管出血）における発生頻度は1.4%，クローン病確診例で7.4%，疑診例では6.3%と報告されている。滞留が発生しても多くの場合腸閉塞など緊急対応を要する状態には陥らないが、自然排出されない場合には、バルーン内視鏡ないし外科手術による回収が試みられる。

パテンシーカプセル（崩壊性疑似カプセル）



Figure 1 PillCam® パテンシーカプセルの構造と消化管開通性判定（コヴィディエン社より提供）

パテンシーカプセルの開通性判定方法

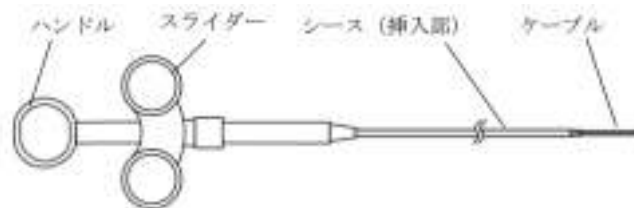
嚥下後経過時間 (T) (単位：時間)	目視 (排便)	X線検査等	PillCam® パテンシーカプセル位置	開通性
T < 30	確認※	—	—	あり
30 ≤ T ≤ 33	確認※	—	—	あり
		実施 (体内に有)	大腸到達※	あり

— : 非該当

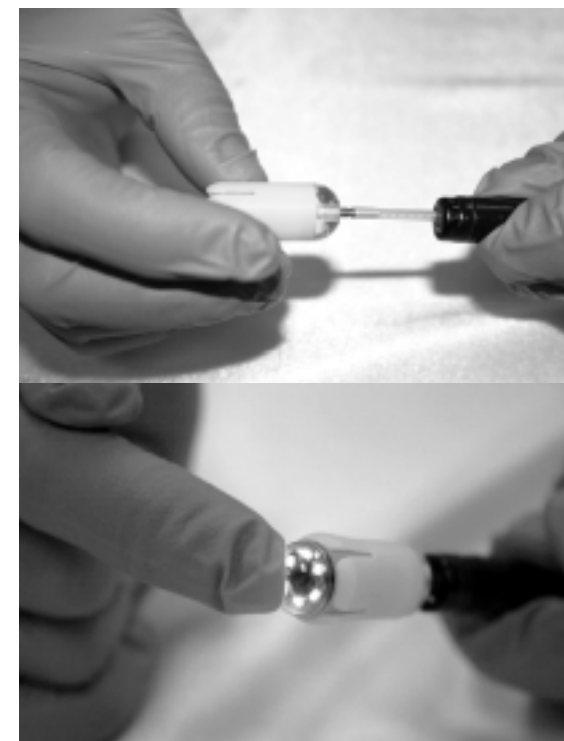


大腸内でパテンシーカプセルが
原形で確認されたX線画像

アドバンス内視鏡挿入補助具®



(2) カプセルカッパ

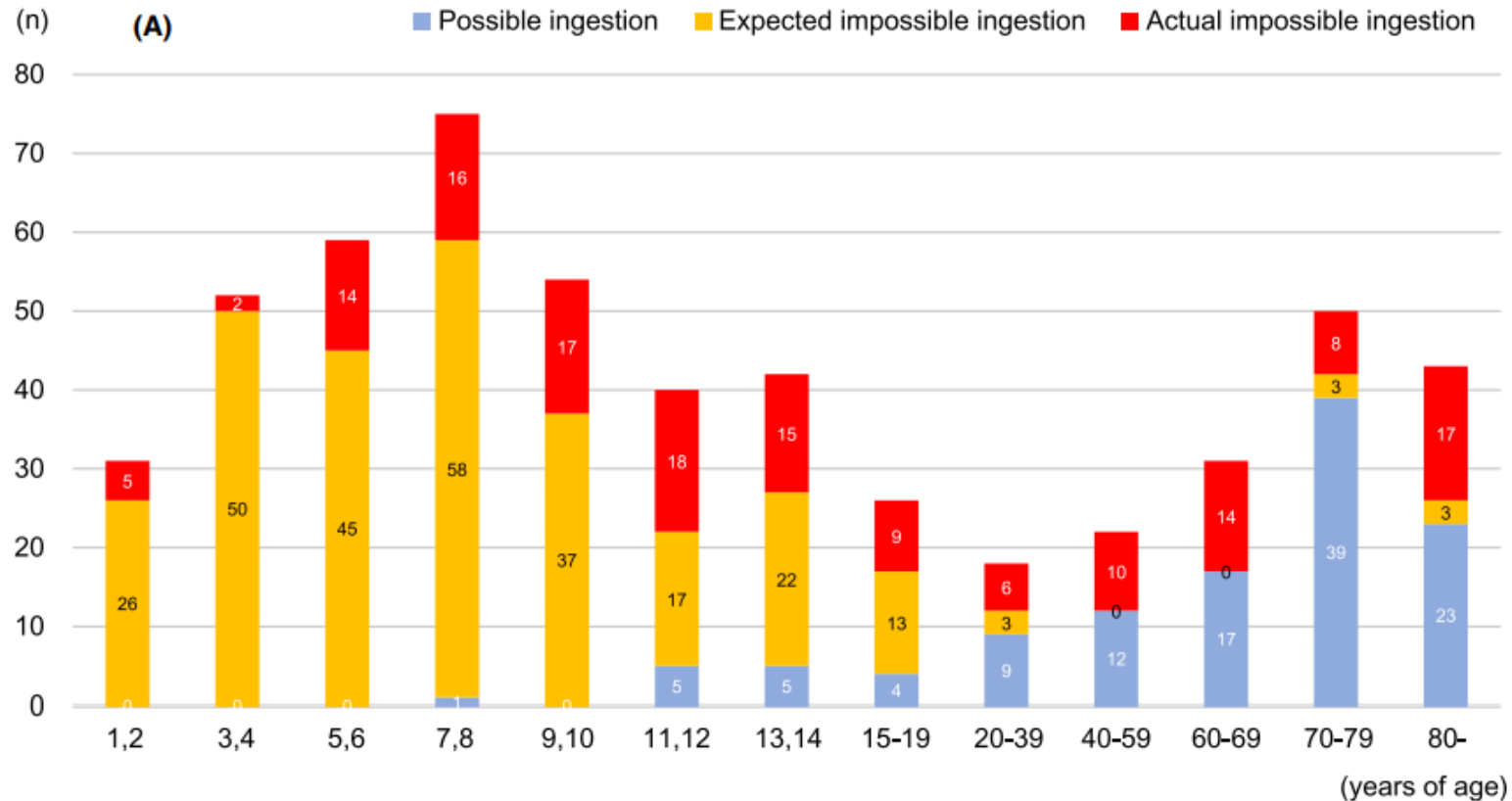


アドバンス内視鏡挿入補助具®取り扱い説明書一部改変

Figure 1 The AdvanCE capsule endoscope delivery device.

Safety and efficacy of the endoscopic delivery of capsule endoscopes in adult and pediatric patients: Multicenter Japanese study (AdvanCE-J study)
Digestive Endoscopy 2021 Naoki Ohmiya et al.

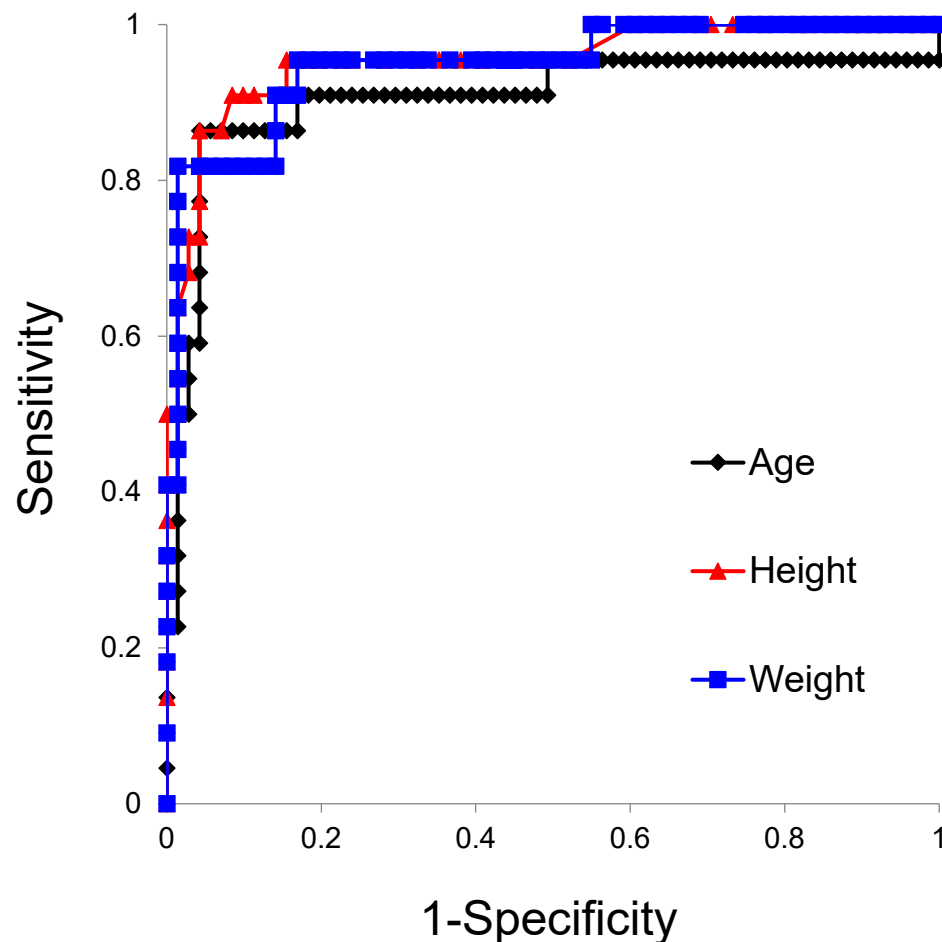
年齢別 カプセル嚥下困難であったり、 嚥下困難が予想された患者数



Safety and efficacy of the endoscopic delivery of capsule endoscopes in adult and pediatric patients:
Multicenter Japanese study (AdvancE-J study) Digestive Endoscopy 2021 Naoki Ohmiya et al.

16未満でカプセル内視鏡検査が施行された93症例の解析

カプセル内視鏡の自己嚥下が困難な予測因子



身長 ; 132 c m以下
体重24,8Kg未満
年齢9歳 2 か月未満

バルーンアシスト内視鏡

バルーンアシスト内視鏡開発

Total enteroscopy with a nonsurgical steerable double-balloon method

Hironori Yamamoto, MD, Yutaka Sekine, MD, Yukihiro Sato, MD, Toshihiko Higashizawa, MD, Tomohiko Miyata, MD, Satoru Iino, MD, Kenichi Ido, MD, Kentaro Sugano, MD

Background: Deep insertion of an enteroscope by use of a push technique is difficult. A new method of enteroscopy was developed, a double-balloon method, to improve the access to the small intestine.

Methods: The new method uses 2 balloons, one attached to the tip of the endoscope and another at the distal end of an overtube. By using these balloons to grip the intestinal wall, the endoscope can be inserted further without forming redundant loops in the small intestine. This method was tried with a standard upper endoscope in 3 patients and with a longer enteroscope in 1 patient.

Results: Despite its short length the upper endoscope was successfully inserted as far as 30 to 50 cm beyond the ligament of Treitz in the 3 patients. In the fourth patient the longer enteroscope was successfully inserted beyond the ileo-cecal valve.

Conclusions: The double-balloon method facilitates endoscopic access to the small intestine.

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From the Department of Gastroenterology, Jichi Medical School, Tochigi, Japan.

Presented in part at the annual meeting of the American Society for Gastrointestinal Endoscopy, May 2000, San Diego, California.

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doi:10.1067/mge.2001.112181

Although the quest to inspect the mucosal surface of the small intestine began early in the fiberoptic era, insertion techniques currently in use are far from ideal. The most frequently used method is push enteroscopy. However, deep insertion of an enteroscope is difficult with this method because the force applied to advance the instrument is diminished by the tortuous small intestine. Moreover this causes tremendous discomfort to the patient. Therefore, even with a new-generation video-push enteroscope along with an overtube, the depth of insertion is at most 160 cm beyond the ligament of Treitz, which is far less than the length of the small intestine, which is approximately 14 feet (430 cm).¹⁻³

At present, intra-operative enteroscopy is the most reliable procedure for total visualization of the small intestine.^{2,4} In this method, passage of the enteroscope is assisted by the surgeon during open laparotomy. Although the technique is not difficult, an endoscopist and surgeon are needed to perform the procedure. The biggest disadvantage of this method, however, is its invasive nature.

Nonsurgical total small bowel enteroscopy has been successfully accomplished by 2 different methods,³ one using an instrument pushed over a previously passed guide-string⁵⁻⁷ whereas the other relies on peristalsis to propel a long, flexible fiberoptic endoscope through the intestine.^{8,9} Both methods are tedious, uncomfortable, and time consuming. The guide-string for the "ropeway" method may take several days to pass from mouth to rectum.³ Stretching of the string can damage the small bowel mucosa and cause tremendous pain and discomfort that necessitates the use of general anesthesia. The limitations of

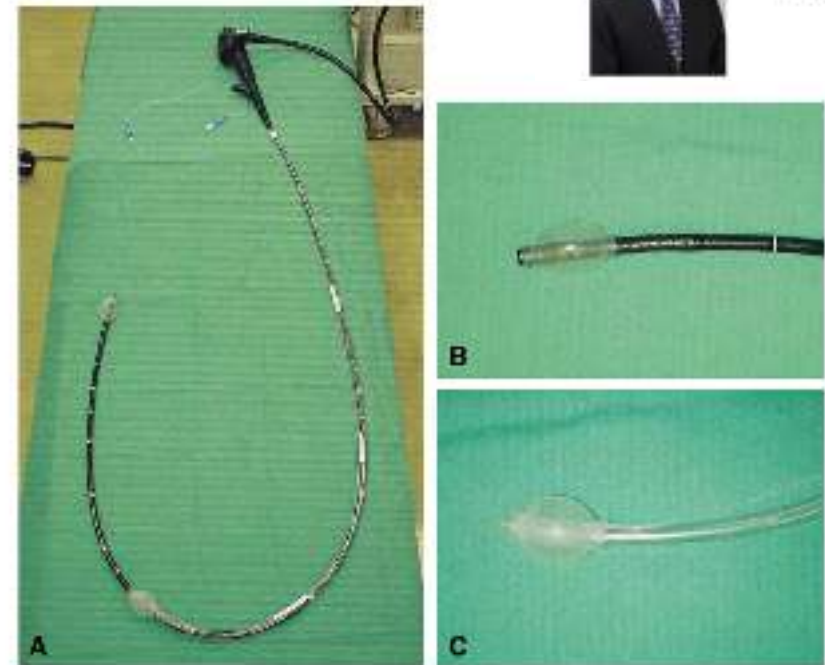


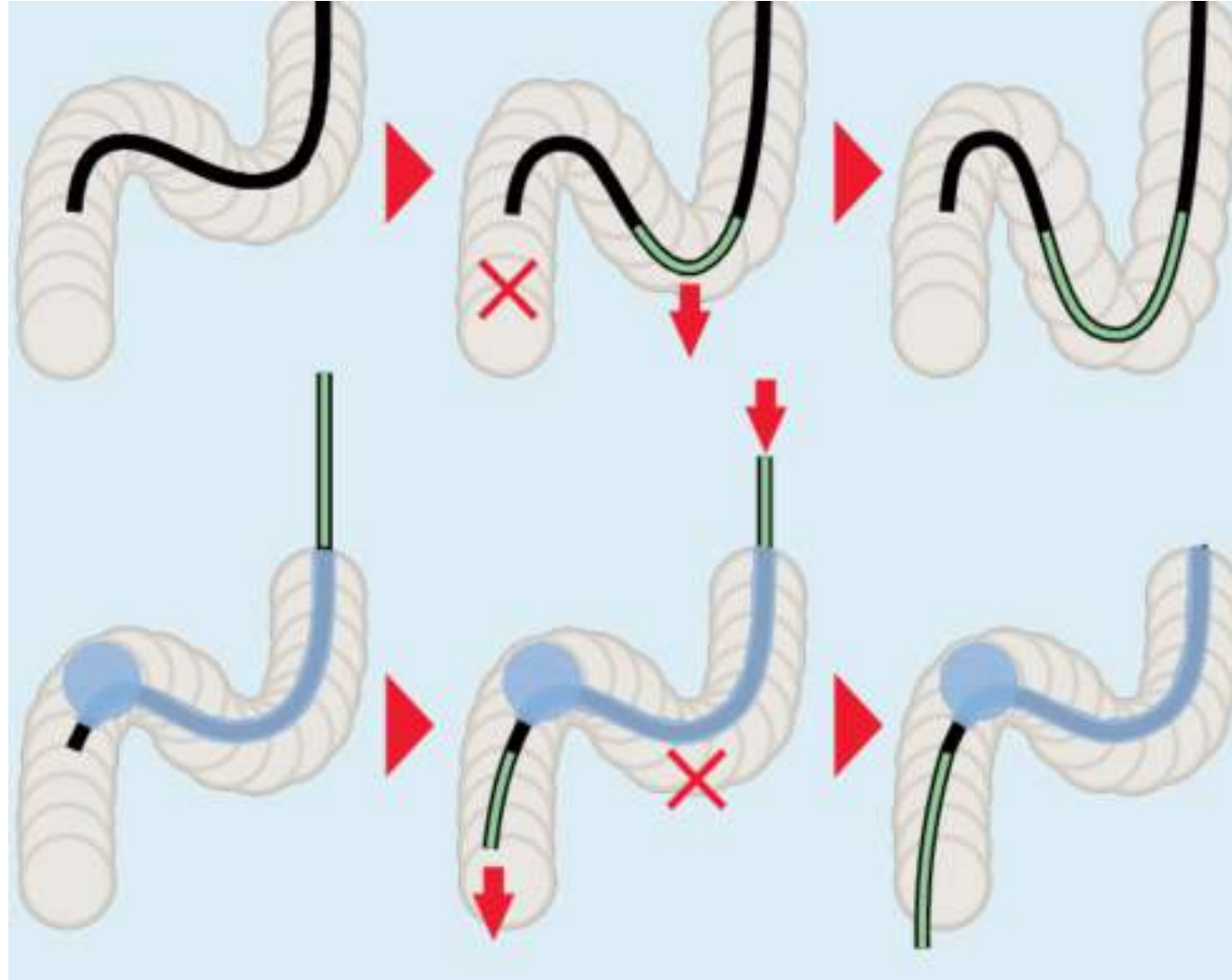
Figure 1. A, Video endoscope (SF-Q200, Olympus) with balloons attached at tip and handmade overtube with another balloon attached in place on the insertion tube. B, Cylindrical balloon attached at tip of enteroscope. C, Distal end of overtube with attached balloon.



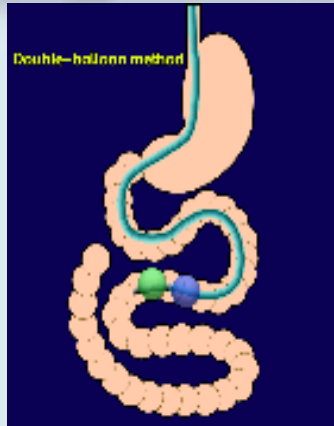
消化器科内科学専門医
山本博徳

Total enteroscopy with a nonsurgical steerable double-balloon method. *Gastrointest Endosc.* 2001

バルーンアシスト内視鏡 挿入原理



バルーンアシスト小腸内視鏡検査



DOUBLE BALLOON SYSTEM

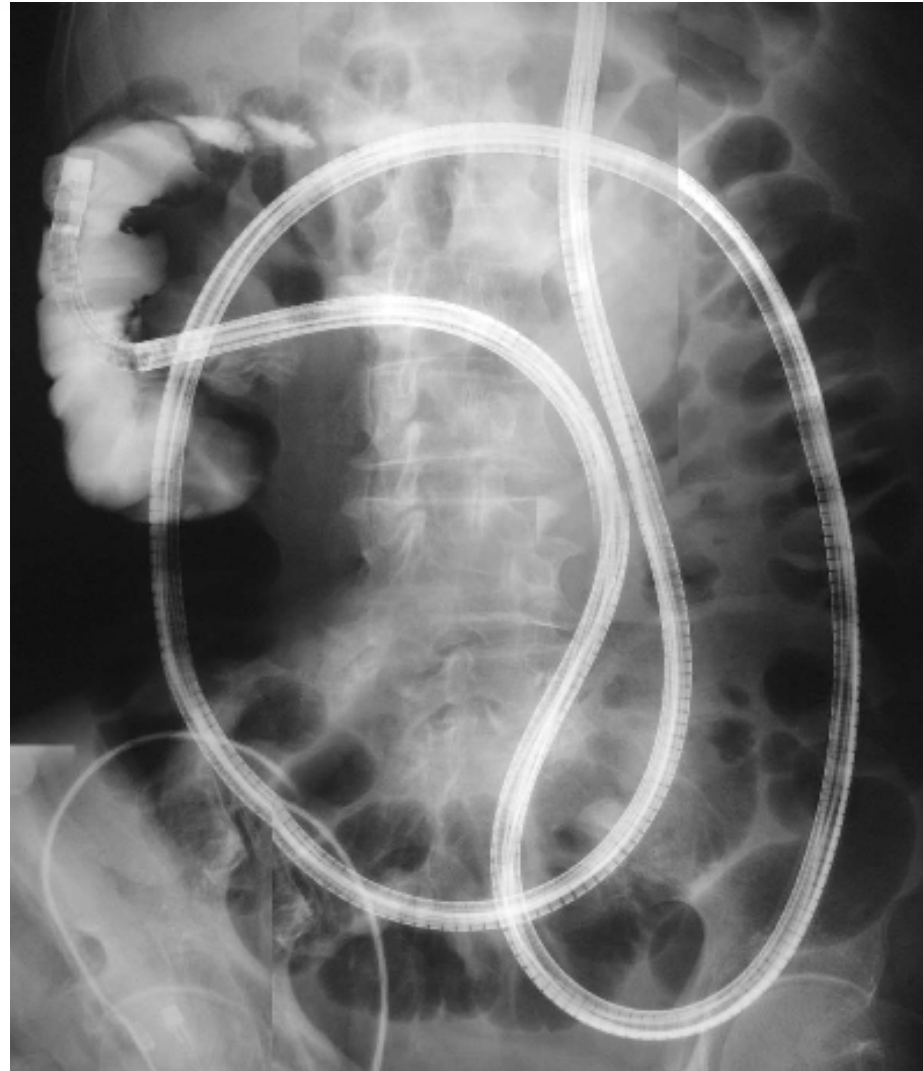


小腸用 細径スコープ EN-50P5/20
汎用処置スコープ EI-530B



オーバチューブ TS-12140/TS-13140

バルーンアシスト小腸内視鏡検査 経口的挿入による盲腸到達時のX線画像



本日の内容

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- 小腸疾患の診断と治療 各論
- 新しい小腸疾患診療の展開

小腸内視鏡の適応疾患

血管性疾患

血管異形成
Dieulafoy's lesion
動静脈奇形
Osler-Weber-Rendu 病（遺伝性出血性毛細血管拡張症）
Blue rubber bleb nevus syndrome
静脈瘤

腫瘍性疾患

腺腫
原発性小腸癌
転移性小腸癌
悪性リンパ腫
GIST
カルチノイド（neuroendocrine tumor）

Peutz-Jeghers 症候群
Cowden 病
Inflammatory fibroid polyp
Pyogenic granuloma
脂肪腫
血管腫
リンパ管腫

炎症性疾患（非感染性）

クローン病
腸管バーチエット病/単純性潰瘍
CEAS/非特異性多発性小腸潰瘍
CMUSE
虚血性小腸炎
IgA 血管炎/ヘノッホ・シェーンライン紫斑病
Cronkhite-Canada 症候群

好酸球性胃腸炎
放射線性小腸炎
移植片対宿主病（graft versus host disease）
Whipple 病
セリアック病
ループス腸炎
NSAIDs 起因性小腸粘膜傷害

炎症性疾患（感染性）

腸結核
エルシニア腸炎
ウイルス性胃腸炎（ノロウイルス、ロタウイルス）

その他

アミロイドーシス
寄生虫感染
メッケル憩室
異物

**小腸疾患（non-CD）を見直す
—腫瘍・血管性病変・炎症・希少疾患まで—
日本消化器病学会雑誌 2018**

小腸疾患の診断と治療 各論

✓血管性病変

✓腫瘍性病変

✓炎症性疾患（粘膜病変）

✓その他

原因不明消化管出血 OGIB (obscure gastrointestinal bleeding) →small-bowel bleeding(SSB)

原因不明消化管出血(OGIB)

上部及び下部消化管内視鏡検査を行っても原因不明の消化管出血

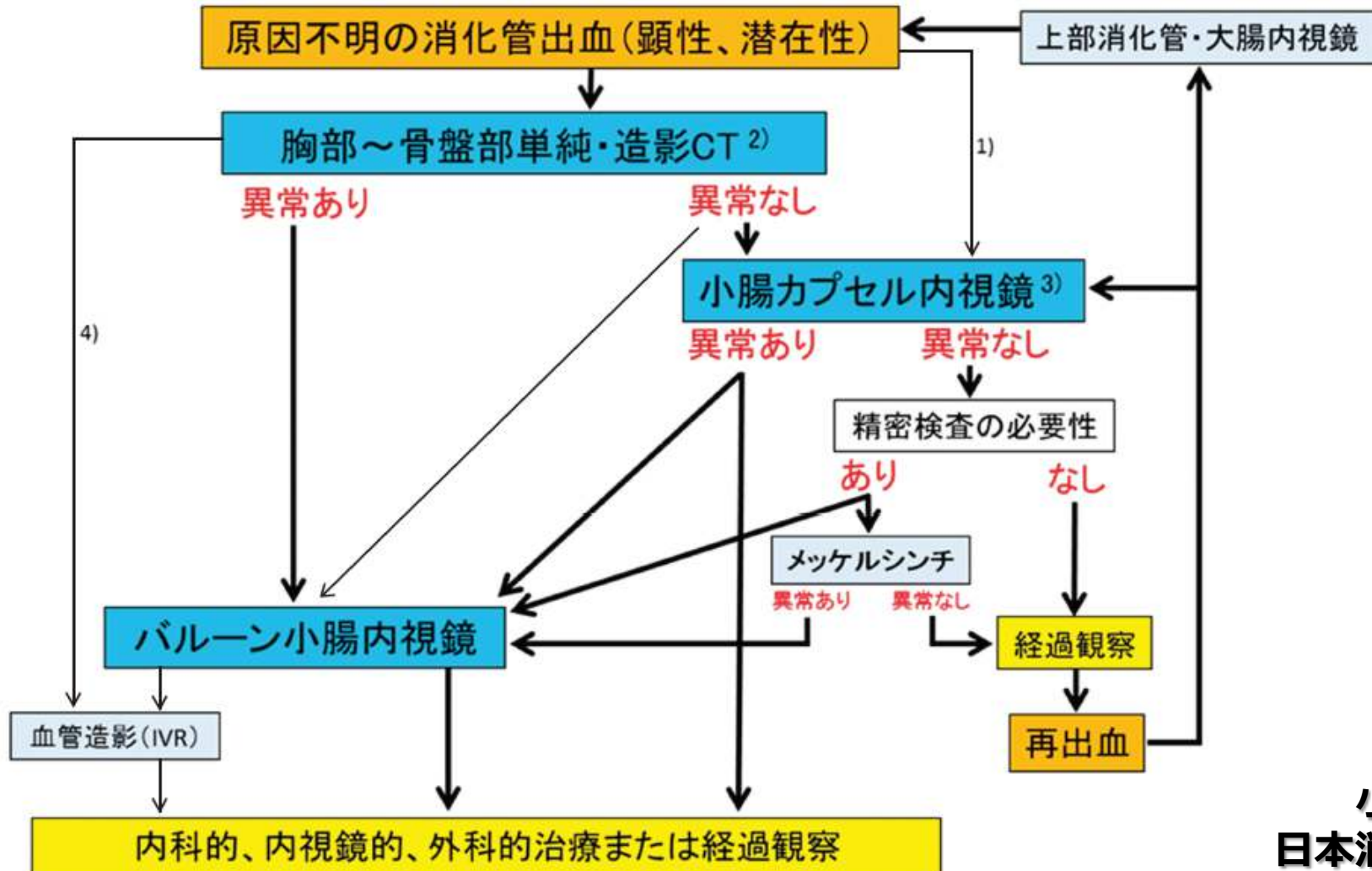
顕在性出血(Overt)

再発または持続する下血や血便などの可視的出血 ① on going ② previous

潜在性出血(Occult)

再発または持続する鉄欠乏性貧血(IDA) および／または
便潜血検査(FOBT)陽性 (但し、大腸内視鏡検査で異常がなく貧血がなければ除外)

原因不明の消化管出血 診断アルゴリズム



OGIB患者にはカプセル内視鏡（CE） をまずは施行する

Obscure gastrointestinal bleeding

ESGE recommends small-bowel video capsule endoscopy as the first-line investigation in patients with obscure gastrointestinal bleeding (strong recommendation, moderate quality evidence).

Small-bowel capsule endoscopy and device-assisted enteroscopy for diagnosis and treatment of smallbowel disorders: European Society of Gastrointestinal Endoscopy (ESGE) Clinical Guideline Endoscopy 2015

Statement : 1-3

小腸カプセル内視鏡は苦痛なく、容易に小腸全域の内視鏡観察ができるため、OGIB のスクリーニングに極めて有用な検査法である。ただし、通過速度の速い十二指腸・上部空腸の病変、カプセル内視鏡が通過しない術後再建腸管の病変や、憩室など見落としがあることを留意すべきである。

Delphi 法による評価 中央値：9，最低値：8，最高値：9

Evidence level：Ⅱ～Ⅴ，推奨度：C1

OGIB患者に対するCEの病変検出率

Table 3. Diagnostic Yield of Capsule Endoscopy According to Type of Bleeding

Type of bleeding	Type of finding (%)		
	Positive	Suspicious	Negative
Overt-ongoing (n = 26)	24 (92.3)	0 (0.0)	2 (7.7)
Overt-previous, overall (n = 31)	4 (12.9)	5 (16.1)	22 (71.0)
10–14 days (n = 3)	2 (66.6)	0 (0.0)	1 (33.3)
3–4 weeks (n = 3)	1 (33.3)	2 (66.6)	0 (0.0)
2–3 months (n = 9)	0 (0.0)	2 (22.2)	7 (77.8)
4–6 months (n = 11)	1 (9.1)	0 (0.0)	10 (90.9)
7–12 months (n = 5)	0 (0.0)	1 (20.0)	4 (80.0)
Occult (n = 43)	19 (44.2)	10 (23.2)	14 (32.6)

**OGIB患者に対
するCE検査
overt症例では
92.3%
occultには
44.2%
で病変検出**

OGIB患者にはCE検査を出血のエピソード後 早期に施行すると病変検出率が高まる

In patients with overt obscure gastrointestinal bleeding ESGE recommends performing small-bowel capsule endoscopy as soon as possible after the bleeding episode, optimally within 14 days, in order to maximize the diagnostic yield (strong recommendation, moderate quality evidence).

Small-bowel capsule endoscopy and device-assisted enteroscopy for diagnosis and treatment of smallbowel disorders: European Society of Gastrointestinal Endoscopy (ESGE) Clinical Guideline Endoscopy 2015

Statement : 1-2

出血から2週間を過ぎると診断能が低くなることが知られている。しかし、2週間以内であっても診断できないこともあるし、内視鏡的に出血を認めても出血源が同定できないことすらあるため、出血源を同定するまで可及的速やかに繰り返し検査することが必要である。

Delphi 法による評価 中央値：9，最低値：8，最高値：9

Evidence level：IVb～V，推奨度：C1

OGIB(overt)患者群において入院後3日以内にCEをした方が、 3日以降にCE施行群より病変検出率が高い

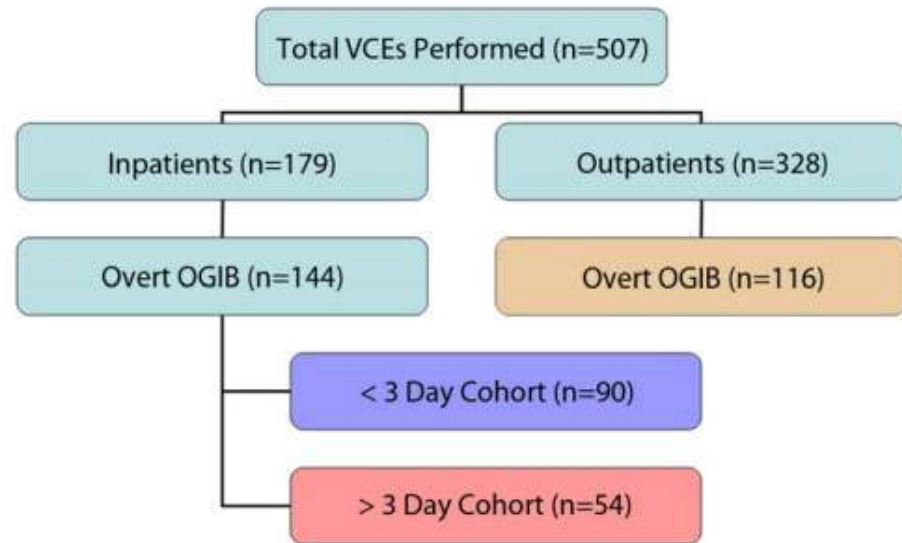


Figure 1. Flow diagram illustrating the study design. *OGIB*, obscure GI bleeding; *VCE*, video capsule endoscopy.

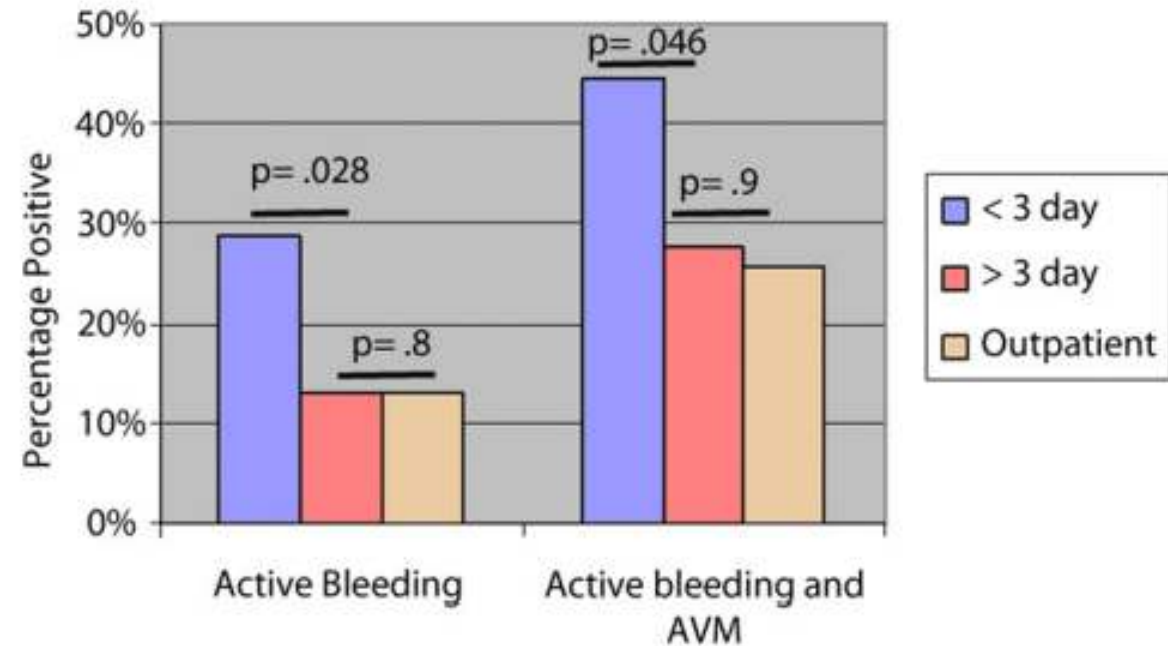


Figure 2. Yield of VCE to detect an active bleed and combined yield to detect an active bleed and arteriovenous malformation for the 3 cohorts.

Timing of video capsule endoscopy relative to overt obscure GI bleeding: implications from a retrospective study
GASTROINTESTINAL ENDOSCOPY 2013

OGIB患者には消化管透視検査や血管造影検査よりCE検査を考慮 しかしCTがCEと相補的検査になりえる患者もいる

ESGE recommends performance of small-bowel capsule endoscopy as the first-line examination, before consideration of small-bowel radiographic studies or mesenteric angiography, when small-bowel evaluation is indicated for obscure gastrointestinal bleeding (strong recommendation, high quality evidence). Computed tomography enterography/enteroclysis may be a complementary examination to capsule endoscopy in selected patients (weak recommendation, low quality evidence).

Small-bowel capsule endoscopy and device-assisted enteroscopy for diagnosis and treatment of smallbowel disorders: European Society of Gastrointestinal Endoscopy (ESGE) Clinical Guideline Endoscopy 2015

Statement : 1-5

内視鏡的止血術の適応は、血管性病変の他、露出血管を伴う潰瘍・腫瘍・憩室からの出血などである。方法は、上部・下部消化管内視鏡で用いられるものと同様である。ただし、内視鏡的止血術が困難な場合は、外科的治療やIVR（interventional radiology）などの適応となる。

Delphi 法による評価 中央値：9，最低値：8，最高値：9
Evidence level：IVb～VI，推奨度：C1

各種消化管出血精査検査 前向き観察研究で出血源が同定できた症例割合

Lesions with a high probability of bleeding (%)

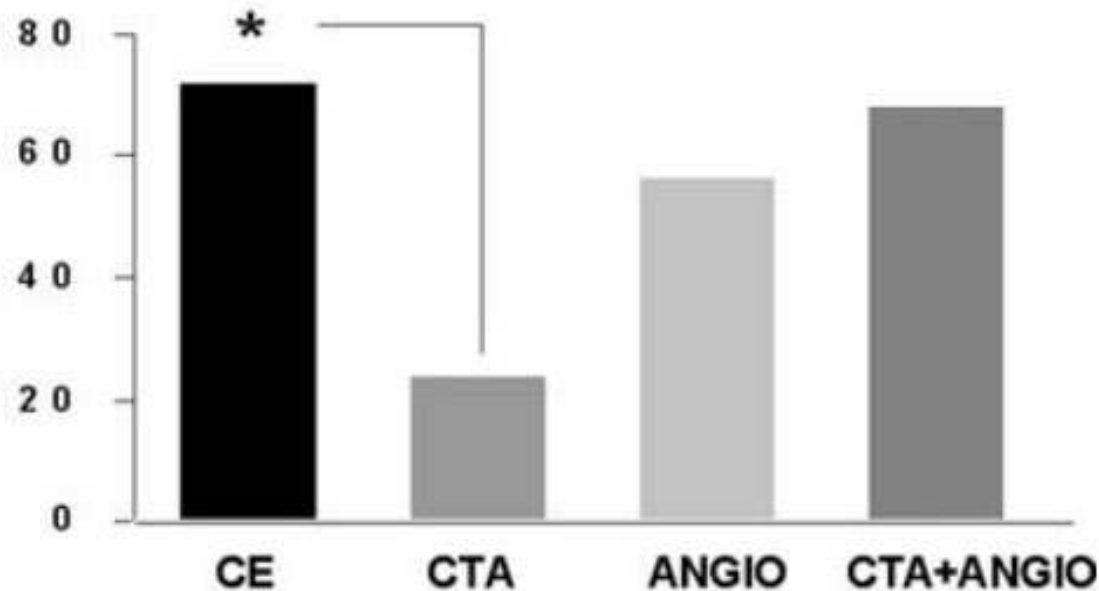


Figure 3. Diagnostic yield of each technique in patients with OGIB.
* $P < 0.01$ versus CT angiography.

CE; 72%
(50.6-87.9)
CT-angiography; 24%
(9.4-45.1%)
Angiography; 56%
(34.9-75.6%)

Capsule Endoscopy Versus Computed Tomographic or
Standard Angiography for the Diagnosis of Obscure
Gastrointestinal Bleeding
American Journal of Gastroenterology 2015

Overt OGIB患者にはバルーンアシスト内視鏡に関しても 診断及び治療も可能なのではじめに施行することも考慮される

ESGE suggests that emergency small-bowel capsule endoscopy should be considered in patients with ongoing overt obscure gastrointestinal bleeding (weak recommendation, moderate quality evidence).

In such patients, ESGE suggests that device-assisted enteroscopy should also be considered as a possible first-line test, given that it allows diagnosis and treatment in the same procedure (weak recommendation, low quality evidence).

**Small-bowel capsule endoscopy and device-assisted
enteroscopy for diagnosis and treatment of smallbowel
disorders: European Society of Gastrointestinal Endoscopy
(ESGE) Clinical Guideline
Endoscopy 2015**

Statement : 1-4

バルーン内視鏡は出血部位の同定，色素内視鏡，超音波内視鏡を含めた詳細な観察，さらに生検や内視鏡治療ができる．一方，侵襲性を考慮し偶発症に留意すべきである．

Delphi 法による評価 中央値：9，最低値：8，最高値：9
Evidence level：IVb～VI，推奨度：C1

OGIB患者におけるCE及びDAEによる診断率の差

Table 2 Summary of diagnostic yields

Study	n	Positive CE	Positive DBE	DBE yield after prior positive CE	DBE yield after prior negative CE
Matsumoto 2005	75	10 (13.3%)	8 (10.7%)	—	—
Hadithi 2006	75	28 (37.3%)	21 (28.0%)	20/28 (71.4%)	7/21 (33.3%)
Mehdizadeh 2006	115	63 (54.8%)	57 (49.6%)	41/63 (65.1%)	16/52 (30.8%)
Nakamura 2006	28	9 (32.1%)	12 (42.9%)	9/17 (52.9%)	3/11 (27.3%)
Fujimori 2007	45 (39 [†])	18 (40.0%)	18 (100%)	6/18 (33.3%)	2/20 (10%)
Ohmiya 2007	74	37 (50.0%)	39 (52.7%)	—	—
Kameda 2008	32	13 (40.6%)	21 (65.6%)	1/23 (4.3%)	2/3 (66.7%)
Arakawa 2009	74	10 (13.5%)	34 (46.1%)	3/10 (30.0%)	11/34 (32.4%)
Fukumoto 2009 [‡]	42	16 (38.1%)	18 (42.9%)	—	—
Marmo 2009	193	74 (38.3%)	132 (68.4%)	124/134 (92.5%)	8/19 (42.1%)

[†]45 patients had CE but only 36 patients underwent DBE in the study by Fujimori et al.

[‡]Unpublished data from a subgroup of patients with OGIB, provided by personal communication with study author Akira Fukumoto, MD PhD, Onomichi, Japan.

CE, capsule endoscopy; DBE, double balloon enteroscopy; OGIB, obscure GI bleeding; —, reclassification or subgroup results could not be discerned from the data.

OGIB患者におけるCEによる診断率は
61.7% (47.3-76.1)、
DAEによる診断率は
55.5% (48.9-62.1)

Double balloon enteroscopy and capsule endoscopy for
obscure gastrointestinal bleeding: An updated meta-analysis
Journal of Gastroenterology and Hepatology 2011

OGIB患者におけるDAEとCEによる 診断率は違いはない

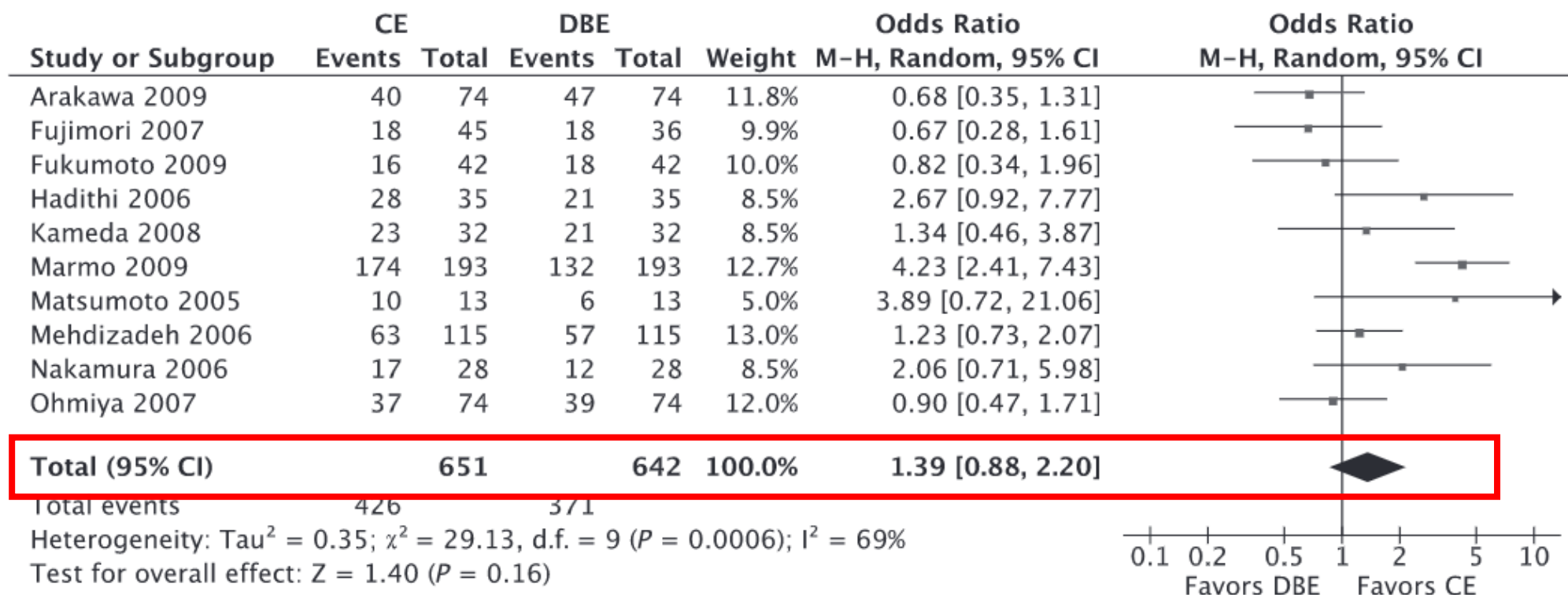


Figure 2 Relative diagnostic yield of capsule endoscopy (CE) compared with double balloon enteroscopy (DBE). CI, confidence interval; d.f., degrees of freedom.

**Double balloon enteroscopy and capsule endoscopy for
obscure gastrointestinal bleeding: An updated meta-analysis**
Journal of Gastroenterology and Hepatology 2011

小腸出血の内視鏡肉眼分類 矢野・山本分類

Type 1a:		Punctate erythema (less than 1 mm) with or without oozing
Type 1b:		Patchy erythema (a few mm) with or without oozing
Type 2a:		Punctate lesions (less than 1 mm) with pulsatile bleeding
Type 2b:		Pulsatile red protrusion without surrounding venous dilatation
Type 3:		Pulsatile red protrusion with surrounding venous dilatation
Type 4:		Other lesions not classified into any of the above categories

Figure 1. Endoscopic classification of small-intestinal vascular lesions (Yano-Yamamoto classification).

- (1) 静脈・毛細血管の特徴を持つ病変 (angiodysplasia) で一番頻度が多く、argon plasma coagulation(APC)や高周波焼灼止血がされるType1a、1b病変
- (2) 動脈の特徴を持つ病変 (Dieulafoy's lesion) で、クリップ止血されるType2a、2b病変
- (3) 動脈と静脈の特徴を持つ病変 (arteriovenous malformation; AVM) である、Type3 クリップや塞栓術もしくはは外科治療が施行される

症例 NSAIDs起因性小腸潰瘍

症例 ; 60歳台の男性

現病歴 ; 濾胞性リンパ腫、頸椎症にて当院通院中。定期外来にて貧血の進行指摘されて当科紹介受診。ヒトヘモ強陽性。GIF: S S B E 所見のみ、CF: 回腸末端に一部浅い潰瘍多発。小腸精査目的で小腸カプセル内視鏡施行された。 R-CHOP治療中とセレコキシブ（セレコックス®）、ブレガバリン（リリカ®）投薬中であつた。



小腸疾患の診断と治療 各論

✓血管性病変

✓腫瘍性病変

✓炎症性疾患（粘膜病変）

✓その他

OGIBや鉄欠乏性貧血（IDA）に小腸腫瘍症例が存在する CEによる早期発見が重要

Small-bowel tumours

ESGE recommends early use of small-bowel video capsule endoscopy in the search for a small-bowel tumour when obscure gastrointestinal bleeding and iron-deficiency anaemia are not explained otherwise (strong recommendation, moderate quality evidence).

In the setting of suspicion of a small-bowel tumour, ESGE does not recommend specific investigations before small-bowel capsule endoscopy in patients without evidence for stenosis or previous small-bowel resection (strong recommendation, low quality evidence).
ESGE recommends consideration of device-assisted enteroscopy in preference to small-bowel capsule endoscopy if imaging tests have already shown suspicion of small-bowel tumour (strong recommendation, low quality evidence).

**明かな狭窄や術後症例でなければまずはCE施行するが、
病変があればDAE検査が考慮される。**

小腸腫瘍の存在が疑わしい時は 横断的画像検査（CT検査など）の施行をまずは優先

ESGE recommends cross-sectional imaging to ascertain operability when there is a small-bowel capsule endoscopy finding of small-bowel tumour with a high diagnostic certainty. When there is uncertain diagnosis of small-bowel tumour at capsule endoscopy, biopsy sampling by device-assisted enteroscopy is required (strong recommendation, low quality evidence).

When a submucosal mass is detected by small-bowel capsule endoscopy, ESGE recommends confirmation of the diagnosis by device-assisted enteroscopy (strong recommendation, low quality evidence).

When capsule endoscopy shows high suspicion of submucosal mass and there is a negative but incomplete device-assisted enteroscopy, ESGE suggests cross-sectional imaging tests to confirm the diagnosis (weak recommendation, low quality evidence).

CE検査施行しても、小腸腫瘍存在が不明確な時や、粘膜下腫瘍が疑わしい時はDAE検査を考慮する。

大きな小腸腫瘍は逆にCEでは偽陰性になることに注意

TABLE 6. Diagnostic yields of tumors or polyps (%)

Size	Origin	CECT*	FE*	VCE†	DBE‡
≤10 mm	Epithelial	0/9 (0)	0/2 (0)	6/8 (75)	19/19 (100)
	Subepithelial	5/21 (24)	4/8 (50)	24/29 (83)	32/33 (97)
>10 mm	Epithelial	18/31 (58)§	29/30 (97)	29/32 (91)	41/42 (98)
	Subepithelial	55/60 (92)	38/40 (95)	25/32 (78)	60/64 (94)

CECT, Contrast-enhanced abdominal CT; FE, fluoroscopic enteroclysis; VCE, videocapsule endoscopy; DBE, double-balloon endoscopy or enteroscopy.

* $P < .0001$, tumors ≤10 mm compared with >10 mm.

† $P = .8784$, tumors ≤10 mm compared with >10 mm.

‡ $P = .9214$, tumors ≤10 mm compared with >10 mm.

§ $P = .0003$, epithelial tumors >10 mm compared with subepithelial tumors >10 mm.

Statement : 3-1

大きな小腸腫瘍は小腸カプセル内視鏡で偽陰性のことがあり、初期検査としてはCT、MRIや小腸造影が奨められる。ただし、小さな腫瘍性病変や平坦病変の検出については、小腸カプセル内視鏡の方が優れている。

Delphi 法による評価 中央値：9，最低値：8，最高値：9

Evidence level：I～VI，推奨度：C1

Enteroscopic and radiologic diagnoses, treatment, and prognoses of small-bowel tumors
GASTROINTESTINAL ENDOSCOPY 2012

小腸内視鏡ガイドライン
日本消化器内視鏡学会雑誌 2015

粗大病変描出が強いCTと微小病変描出が強いCEを併用すれば 存在診断率としてはDAEにせまる診断率が得られる

TABLE 7D. Comparison of diagnostic yields of DBE and a combination test (CECT and VCE)*†

	DBE		
	Positive	Negative	Total
CECT + VCE			
Positive	66	1	67
Negative	7	0	7
Total	73	1	74

DBE, Double-balloon endoscopy or enteroscopy; CECT, contrast-enhanced abdominal CT; VCE, videocapsule endoscopy.

* $P = .0771$, odds ratio 7.00, exact McNemar test.

†The diagnostic yield of both CECT and VCE (91%) was not significantly different from that of DBE (99%).

Enteroscopic and radiologic diagnoses, treatment, and prognoses of small-bowel tumors GASTROINTESTINAL ENDOSCOPY 2012

Statement : 3-2

バルーン内視鏡は他の小腸画像検査法に比べ腫瘍の存在診断や質的診断能に優れ、精密検査として奨められる。また、内視鏡治療も可能である。

Delphi 法による評価 中央値 : 9, 最低値 : 8, 最高値 : 9
Evidence level : I ~ V, 推奨度 : C1

小腸内視鏡ガイドライン
日本消化器内視鏡学会雑誌 2015

小腸内視鏡生検、粘膜切除術の注意点 易出血性

Statement : 3-3

内視鏡下生検は確定診断や治療方針決定に有用であるが、GIST や血管系腫瘍などでは、輸血を要する程の大量出血をきたす場合があり、注意が必要である。

Delphi 法による評価 中央値 : 9, 最低値 : 8, 最高値 : 9

Evidence level : IVb~VI, 推奨度 : C1

Statement : 3-4

内視鏡的切除術は、粘膜内にとどまる腫瘍や、粘膜下層に限局する粘膜下腫瘍が適応となる。ただし、易出血性の血管系腫瘍や重積した腫瘍では出血や穿孔の危険性があり、外科的切除も考慮する。

Delphi 法による評価 中央値 : 9, 最低値 : 8, 最高値 : 9

Evidence level : IVb~V, 推奨度 : C1

家族性大腸腺腫症（FAP）には十二指腸や乳頭に腫瘍性病変が多発しやすい 通常の直視もしくは側視鏡による観察が必要

Familial adenomatous polyposis

ESGE recommends that surveillance of the proximal small bowel in familial adenomatous polyposis is best performed using conventional forward-viewing and side-viewing endoscopes (strong recommendation, moderate quality evidence).

When small-bowel investigation is clinically indicated in familial adenomatous polyposis, ESGE suggests that small-bowel capsule endoscopy and/or cross-sectional imaging techniques may be considered for identifying polyps in the rest of the small bowel, but the clinical relevance of such findings remains to be demonstrated (weak recommendation, moderate quality evidence).

Small-bowel capsule endoscopy and device-assisted enteroscopy for diagnosis and treatment of smallbowel disorders: European Society of Gastrointestinal Endoscopy (ESGE) Clinical Guideline Endoscopy 2015

Statement : 3-6

家族性大腸腺腫症は大腸以外の消化管病変として、十二指腸（乳頭部を含む）・空腸を中心とした小腸に上皮性腫瘍が好発する。小腸内視鏡は小腸腫瘍の診断および経過観察に有用である。

Delphi 法による評価 中央値：9，最低値：8，最高値：9

Evidence level：IV，推奨度：C1



Peuts-Jeghers症候群 10-15mm以上のポリープはDBEにて切除していく

Peutz-Jeghers syndrome

ESGE recommends device-assisted enteroscopy with timely polypectomy when large polyps (>10–15 mm) are discovered by radiological examination or small-bowel capsule endoscopy in patients with Peutz-Jeghers syndrome (strong recommendation, moderate quality evidence).

Small-bowel capsule endoscopy and device-assisted enteroscopy for diagnosis and treatment of smallbowel disorders: European Society of Gastrointestinal Endoscopy (ESGE) Clinical Guideline Endoscopy 2015

Statement : 3-5

Peutz-Jeghers 症候群の過誤腫性ポリープは小腸に好発し、小腸内視鏡はその診断や治療、経過観察に有用である。ポリープの内視鏡的切除は、腸重積、出血および癌化などの合併症に対する治療・予防に有効である。

Delphi 法による評価 中央値：9，最低値：8，最高値：9

Evidence level：IVb～V，推奨度：C1

小腸腺癌を発見したら FAPやLynch症候群が隠れていないか検討

Table 1 Genetic risks and familial associations of small bowel carcinoma

Syndrome	Mode of inheritance	Mutated/associated gene	Relative risk (95%CI)	Lifetime risk for SBA	Polyps/pathway
FAP ^[7,29]	Autosomal dominant (AD)	APC	330 (132-681)	3%-5%	Adenoma-carcinoma
HNPCC/ LS ^[13,14,29]	AD	MMR (MSH2, MSH6, MLH1 ,PMS2)	291 (71-681)	1%-4%	Adenoma-carcinoma
PJS ^[31,36,37]	AD	STK11	500 (220-1306)	1.7%-13%	Hamartoma, adenoma-Ca
JPS ^[38,41]	AD	BMPRI1A, SMAD4	Unknown	Unknown	Hamartoma, adenoma-Ca
Crohn's disease	Unknown (genome wide studies have associated 140 loci)	Unknown	30-60 ^[44-49] (15-609)	2.2% after/25 yr	Dysplasia-carcinoma
Celiac disease	Association with HLA-DQ2,HLA-DQ8	Unknown	60-80 ^[61-63] (7-240)	< 1%	Adenoma-carcinoma

FAP: Familial adenomatous polyposis; APC: Adenomatous polyposis coli; HNPCC: Hereditary nonpolyposis colorectal cancer; MMR: Mismatch repair gene; LS: Lynch syndrome; PJS: Peutz-Jeghers syndrome; STK11: Serine threonine kinase; JPS: Juvenile polyposis syndrome; BMPRI1A: Bone morphogenetic protein receptor, type IA; SMAD4: Mothers against decapentaplegic homolog; HLA: Human leukocyte antigen complex.

**Genetic risks and familial associations of small bowel
Carcinoma World J Gastrointest Oncol 2016**

小腸疾患の診断と治療 各論

- ✓血管性病変
- ✓腫瘍性病変
- ✓炎症性疾患（粘膜病変）
- ✓その他

クローン病(CD) 下部消化管内視鏡で回腸末端に病変がない時 CE施行することをすすめる

Crohn's disease

ESGE recommends ileocolonoscopy as the first endoscopic examination for investigating patients with suspected Crohn's disease (strong recommendation, high quality evidence). In patients with suspected Crohn's disease and negative ileocolonoscopy findings, ESGE recommends small-bowel capsule endoscopy as the initial diagnostic modality for investigating the small bowel, in the absence of obstructive symptoms or known stenosis (strong recommendation, moderate quality evidence).

Small-bowel capsule endoscopy and device-assisted enteroscopy for diagnosis and treatment of small bowel disorders: European Society of Gastrointestinal Endoscopy (ESGE) Clinical Guideline Endoscopy 2015

Statement : 4-3

バルーン内視鏡は、クローン病の診断と病状評価に有用である。ただし、侵襲性も考慮して他の画像診断も検討する。また、狭窄に対する内視鏡的バルーン拡張術で手術を回避できる場合がある。小腸カプセル内視鏡は、クローン病の診断と経過観察に有用であるが、疑診例も含め滞留の危険性があり、消化管開通性確認用カプセル（パテンシーカプセル）をあらかじめ施行する必要がある。

Delphi 法による評価 中央値：8，最低値：8，最高値：9

Evidence level：I，推奨度：C1

CDにおいてCEはMREやCTEより高い診断率 及び陰性的中率を示す

Table 2. Sensitivity and Specificity of CE, MRE, and CTE for the Diagnosis of CD in the Terminal Ileum With Ileoscopy and/or Surgery as Gold Standard

	CE (n = 69)	MRE (n = 72)	CTE (n = 73)
Sensitivity (%)	100 (79–100)	81 (58–95)	76 (53–92)
Specificity (%)	91 (79–97)	86 (74–94)	85 (72–93)
PPV (%)	76 (53–92)	71 (49–87)	67 (45–84)
NPV (%)	100 (93–100)	92 (80–98)	90 (78–97)
Prevalence of CD in the terminal ileum with gold standard (n)	16	21	21
False positive (n)	5	7	8
False negative (n)	0	4	5

NOTE. In patients examined with CE, MRE, and CTE, a gold standard assessment of the terminal ileum was obtained in 69, 72, and 73 patients, respectively. 95% confidence intervals are displayed in parentheses.

NPV, negative predictive value; PPV, positive predictive value.

Diagnostic Accuracy of Capsule Endoscopy for Small Bowel Crohn's Disease Is Superior to That of MR Enterography or CT Enterography CLINICAL GASTROENTEROLOGY AND HEPATOLOGY 2011

CDがCEやCTE、MREにおいて疑わしい方にはより 正確な診断が可能なDAEによる精査をすすめる CE及びDAEはCTEやMREの横断的検査とは補完的役割

ESGE recommends device-assisted enteroscopy with small-bowel biopsy in patients with noncontributory ileocolonoscopy and with suspicion of Crohn's disease on small-bowel cross-sectional imaging modalities or small-bowel capsule endoscopy. Device-assisted enteroscopy with small-bowel biopsy is more likely to provide definitive evidence of Crohn's disease than cross-sectional imaging, although the latter offers a useful less invasive alternative that better defines transmural complication (strong recommendation, high quality evidence).

Small-bowel capsule endoscopy and device-assisted enteroscopy for diagnosis and treatment of smallbowel disorders: European Society of Gastrointestinal Endoscopy (ESGE) Clinical Guideline Endoscopy 2015

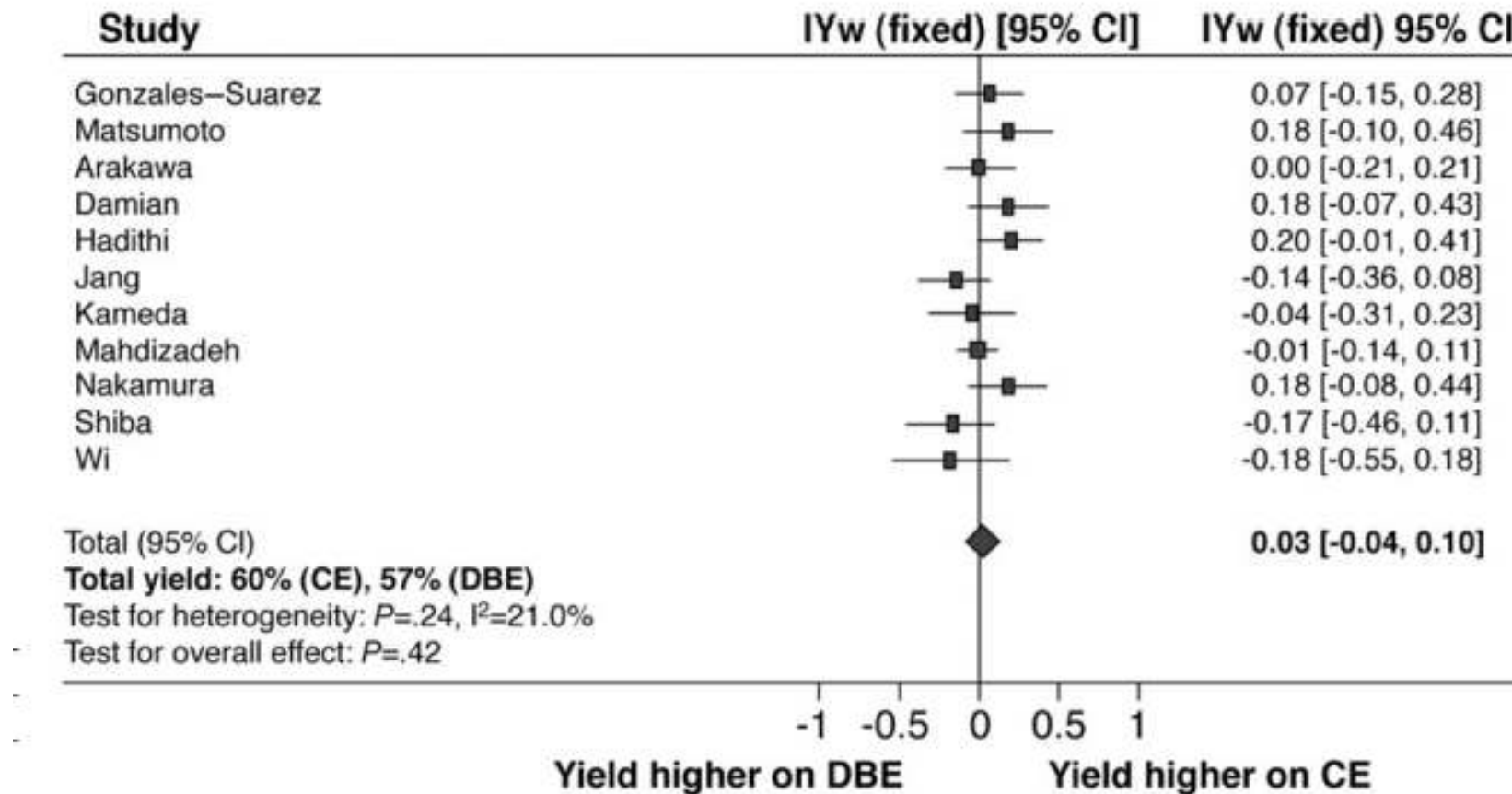
Statement : 4-3

バルーン内視鏡は、クローン病の診断と病状評価に有用である。ただし、侵襲性も考慮して他の画像診断も検討する。また、狭窄に対する内視鏡的バルーン拡張術で手術を回避できる場合がある。小腸カプセル内視鏡は、クローン病の診断と経過観察に有用であるが、疑診例も含め滞留の危険性があり、消化管開通性確認用カプセル（パテンシーカプセル）をあらかじめ施行する必要がある。

Delphi 法による評価 中央値：8，最低値：8，最高値：9

Evidence level：I，推奨度：C1

CDにおいてDBEはCEと同等の診断能



CDの狭窄拡張にDAEは有用

ESGE recommends device-assisted enteroscopy if small-bowel endotherapy is indicated (including dilation of Crohn's disease small-bowel strictures, retrieval of foreign bodies, and treatment of small-bowel bleeding) (strong recommendation, low quality evidence).

Small-bowel capsule endoscopy and device-assisted enteroscopy for diagnosis and treatment of smallbowel disorders: European Society of Gastrointestinal Endoscopy (ESGE) Clinical Guideline Endoscopy 2015

Statement : 4-4

クローン病の小腸病変に対するバルーン拡張術の適応は、狭窄症状ないし口側腸管拡張を有する長軸方向 5 cm 以下の狭窄で、瘻孔、裂溝、膿瘍、深い潰瘍、高度の癒着・屈曲を伴わないものである。偶発症として出血・穿孔などがある。

Delphi 法による評価 中央値：9，最低値：8，最高値：9

Evidence level：I，推奨度：B

CDのバルーン拡張の手技成功率は90%程度 症状改善率は70%程度

Table 1 | Study demographics, frequency of population size, strictures and dilatations

Author	Country	Study design	Population size	Number of strictures (s)	Number of dilatations (d)	Symptomatic response occurrence, S	Technical response occurrence, E	Complication occurrence, C	Study outcome
Total			1089	790	2664				
Pooled unweighted event rate (%)						393/615 (63.9)	403/435 (93)	25/564 (4)	

RCT, randomised control trials; R = Retrospective; P = Prospective; NS = not specified Study Outcome Measures, S = symptomatic response, E = Endoscopic/Technical response, C = Complication, s = number of strictures, d = number of dilatations.

Systematic review with meta-analysis: endoscopic balloon dilatation for Crohn's disease strictures
Aliment Pharmacol Ther 2015

CD バルーン拡張

24か月以内の再拡張率 73.8%・手術率 42.9%

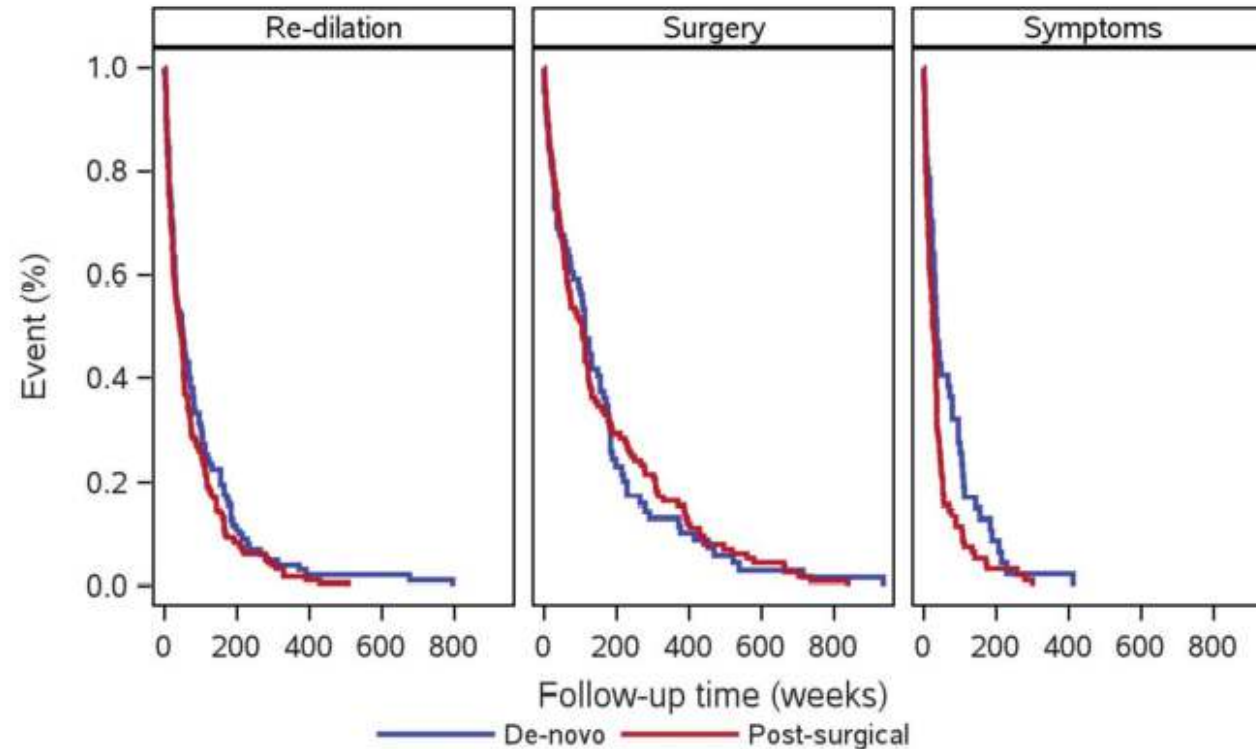


FIGURE 1. Risk of de novo versus anastomotic strictures for redilation, surgery, or symptomatic recurrence during follow-up. Evaluating the future course of de novo strictures (purple lines) and anastomotic strictures (red lines) after endoscopic dilation indicates that the risk for redilation, surgery, and symptomatic recurrence is comparable for both types of strictures during follow-up.

クローン病の内視鏡的バルーン拡張適応症例

狭窄症状ないし口側腸管拡張傾向を有する腸軸方向5cm以下の狭窄で、瘻孔、裂溝、膿瘍、深い潰瘍、高度の癒着・屈曲を伴わないもの

BQ 2 : 長い狭窄に対して EBD は行ってよいか？

Statement : 狭窄が長いほど、拡張の治療成績は悪い。狭窄長 4 ないし 5 cm 以下の病変がよい適応と考えられる。

BQ 1 : 炎症や潰瘍を伴う狭窄に対して EBD は行ってよいか？

Statement : 炎症や潰瘍を伴う狭窄に対する EBD は禁忌ではない。しかし、深い潰瘍は避けるべきである。

FRQ 1 : 膿瘍や瘻孔を伴う狭窄は EBD を行ってよいか？

Statement : 膿瘍や瘻孔を伴う狭窄は EBD の対象としない。

FRQ 2 : 他に EBD の適応から除外すべき狭窄は？

Statement : 癒着を伴う強い屈曲を伴う狭窄は、狭窄部だけでなく、狭窄の前後で穿孔が起きるリスクに注意する。

2021年 クローン病小腸
狭窄に対する内視鏡的バ
ルーン拡張術ガイドライン
(小腸内視鏡診療ガイドラ
イン追補)
日本消化器内視鏡学会雑誌

CDのバルーン拡張治方法

12-15mm・30秒から2分・複数個所同時拡張可能

CQ 3：適切な拡張径は何 mm か？

Recommendation：拡張径は安全性の観点から狭窄径に応じて調整する。有効性の観点から最終目標は一般に 12~15 mm とすることを推奨する。

修正 Delphi 法による評価：中央値 9，最低値 7，最高値 9

推奨の強さ：2，エビデンスレベル：C

FRQ 4：何分間バルーンを拡張状態に保つか？

Statement：一般的には、拡張時間は 30 秒から 2 分 で実施されている。

CQ 2：連続で拡張する狭窄の数に制限はあるか？

Recommendation：手技的に可能であれば、狭窄の 個数による制限なく、拡張することを推奨する。

修正 Delphi 法による評価：中央値 9，最低値 7，最高値 9

推奨の強さ：2，エビデンスレベル：C

CDのバルーン拡張方法

吻合部狭窄も拡張可・抗菌薬予防投与なし・再狭窄予防無効

CQ 1：吻合部狭窄（anastomotic lesion）と非吻合部狭窄（de novo lesion）に対する効果・安全性は同等か？

Recommendation：両者に対する EBD の効果・安全性はほぼ同等であり、いずれにも行うことを推奨する。

修正 Delphi 法による評価：中央値 9，最低値 8，最高値 9

推奨の強さ：2，エビデンスレベル：C

FRQ 5：偶発症への対策として予防的抗菌薬の投与は有効か？

Statement：偶発症への対策として、予防的に抗菌薬を投与する有効性は明らかではない。

FRQ 3：EBD 後の再狭窄の予防として、ステロイドや抗 TNF α 抗体製剤の局所投与は有効か？

Statement：再狭窄の予防策としてのステロイドまたは抗 TNF α 抗体製剤の局所投与の有効性は明らかでない。

CDのバルーン拡張方法

ステントの有効性・予防的バルーン拡張の有効性は不明

FRQ 6：他の手段（内視鏡的狭窄切開術、ステント留置術）の適応は？

Statement：内視鏡的バルーン拡張術以外の内視鏡治療（内視鏡的狭窄切開術、ステント留置術）の有効性・安全性は確立されていない。

FRQ 7：一度 EBD を行った狭窄に対して、無症状でも予防的 EBD を行うか？

Statement：予防的 EBD を行うことの有効性についてのエビデンスは未集積であり、今後の検討課題である。

非特異的小腸潰瘍症アトラス chronic enteropathy associated with SLC02A1 (CEAS)

非特異性多発性小腸潰瘍症 画像診断アトラス

日本医療研究開発機構委託研究開発費(難治性疾患実用化研究事業)
「難治性小腸潰瘍の診断法確立と病態解明に基いた治療法探索」

2016年3月作成

主要所見

A. 臨床的事項

- 1) 複数回の便潜血陽性
- 2) 長期にわたる小球性低色素性貧血と低蛋白血症

B. X線・内視鏡所見(※1)

- 1) 近接、多発する非対称性狭窄、変形(X線所見)
- 2) 近接多発し、境界鮮鋭で浅く斜走、横走する地図状、テープ状潰瘍(内視鏡所見)

C. 切除標本上の特徴的所見(※2)

- 1) 回腸に近接多発する境界鮮鋭で平坦な潰瘍またはその瘢痕
- 2) 潰瘍は地図状ないしテープ状で、横走、斜走する
- 3) すべてUL-IIまでにとどまる非特異性潰瘍

鑑別疾患

- 1) 腸結核(疑診例を含む)
- 2) クローン病
- 3) 腸管ベーチェット病/単純性潰瘍
- 4) 薬剤性腸炎

確診例:

1. 主要所見のAに加え、Bの1)あるいは2)またはCが認められるもの。
2. 十分に検索された標本上Cを満足するもの。

疑診例: 主要所見Aが認められるが、BまたはCの所見が明確でないもの。

非特異的小腸潰瘍症 CEAS（自験例）

● 内視鏡的拡張術施行例画像(疑診例)

非特異性多発性小腸潰瘍症の狭窄に対するバルーン拡張術

非特異性小腸潰瘍症の病態は、ULI-Iと深く、狭窄も緩衝を登き込まない様様狭窄となる。このため、狭窄部に潰瘍を伴っている、大径のバルーン拡張術が比較的安全に行える。ただし、強い屈曲部では拡張用バルーンがまっすぐな棒状になって狭窄前後の腸管壁が穿孔する可能性があるため、充分な長さのガイドワイヤを挿入し、狭窄部前後の腸管形状を整えて行う必要がある。

症例1 50歳代、女性。発症年齢8歳。小腸粘膜SLCO2A7蛋白発現異常(未施行)、SLCO2A7遺伝子変異(未施行)。ヘモグロビン値 6.1g/dL、血清アルブミン値 2.8g/dL。



小腸切開検査

a) 41歳時に剖開する狭窄部に対しての腸管分切部

バルーン内視鏡

b) カットフードを挿入したバルーン内視鏡

バルーン小腸内視鏡検査

c) 拡張下部の炎症3-8mmの狭窄

d) バルーン拡張

<拡張結果>

カットフード(e)を挿入したバルーン小腸内視鏡で、拡張下部の炎症3-8mmの狭窄(6)に対し内視鏡的拡張術を施行した。挿入時に3.5mmまでバルーン拡張ししながら拡張し、戻りながら全体的に狭窄を15mmまでバルーン拡張した。



症例2

20歳代、男性。発症年齢25歳。小腸粘膜SLCO2A7蛋白発現異常(未施行)、SLCO2A7遺伝子変異(未施行)、肥厚性皮膚骨膜症(不明)、ヘモグロビン値 7.8g/dL、血清アルブミン値 4.4g/dL。



カプセル内視鏡検査

a) 19

カプセル内視鏡にでも狭窄は正常な時間が見られ、狭窄部は狭窄を伴うが、狭窄が拡張下部に全長、狭窄をともないカプセル内視鏡が確認し、狭窄部は狭窄のバルーン小腸内視鏡にて確認。

バルーン小腸内視鏡検査

c) 拡張下部に強い炎症性狭窄部を伴う狭窄部(6)を認める(バルーン拡張前)。

d) バルーン拡張後

e) 狭窄カプセル内視鏡のバルーン小腸内視鏡検査で、狭窄部を拡張後、確認したカプセル内視鏡を確認した。



確立された
治療方法はない

同様の症状を呈する家系を詳しく検討することで 発見された (CEAS)

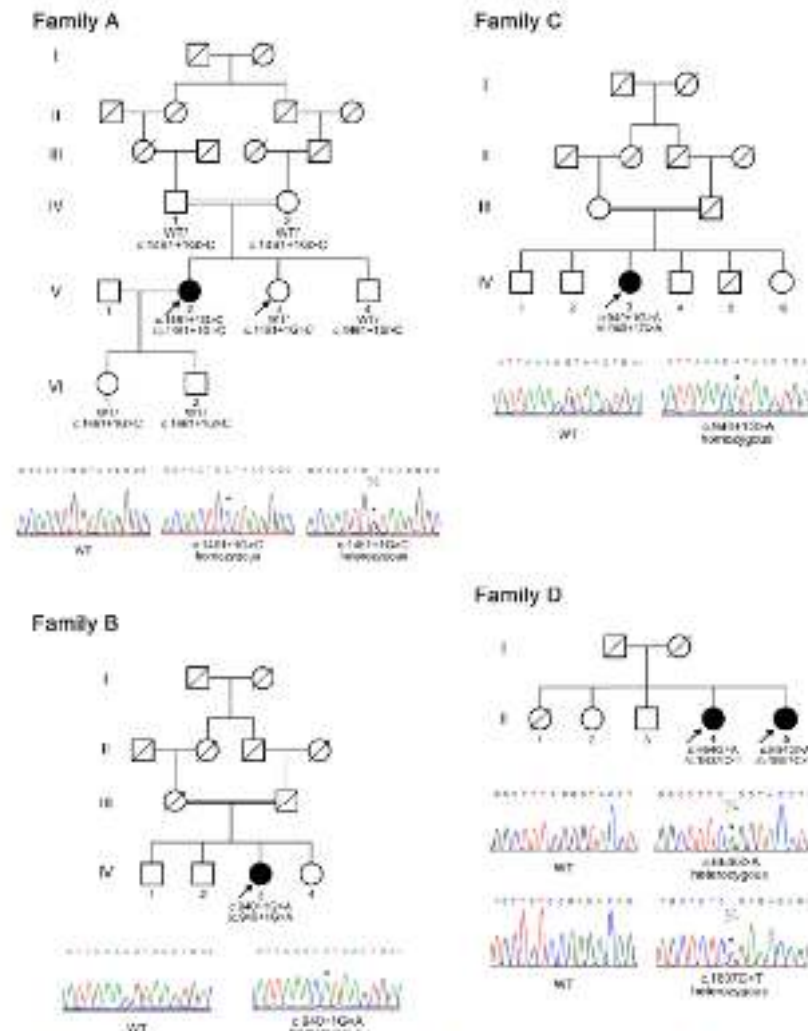


Fig 1. Pedigrees of the families with chronic nonspecific multiple ulcers of the small intestine. The coinheritance of the *SLCO2A1* mutations c.1481+1G>C (exon 10, family A), c.1481+1G>C (exon 10, families B and C), c.664G>A (G222R, family D), and c.1807C>T (R603X, family D) is indicated. Squares represent male family members, circles represent female family members, black symbols represent clinically affected family members, and slashes represent deceased family members. Arrows indicate individuals whose DNA was analyzed by whole-exome sequencing. WT denotes wild-type.

doi:10.1371/journal.pgen.1005581.g001

Table 2. *SLCO2A1* mutations in individuals with chronic nonspecific multiple ulcers of the small intestine.

No.	Genomic Position chr3 (hg19)	Exon	Nucleotide Change	Predicted Effect*	Mutant Allele Frequency			
					CNSU† (n = 16)	Control (n = 747)	HGVD	CD‡ (n = 603)
1	133,674,014	4	c.421G>T [‡]	E141X	-	1/32	1/858	1/1194
2	133,673,888	4	c.547G>A [§]	G183R Deleterious	0/32	0	1/858	1/1194
3	133,672,567	5	c.664G>A	G222R Deleterious	4/32	0	1/858	0
4	133,667,736	7	c.940+1G>A	Splice	-	19/32	3/1494	11/1206
							1330	
5	133,664,028	10	c.1372G>T	V458F Deleterious	2/32	0	0	0
6	133,663,938	10	c.1461+1G>C	Splice	-	2/32	0	0
7	133,654,625	13	c.1807C>T	R603X	-	4/32	0	0

*Mutation pathogenicity according to SIFT, PolyPhen-2, and PROVEAN.

†Initial diagnosis.

‡The variant differs from rs148547180 (chr3:133674014).

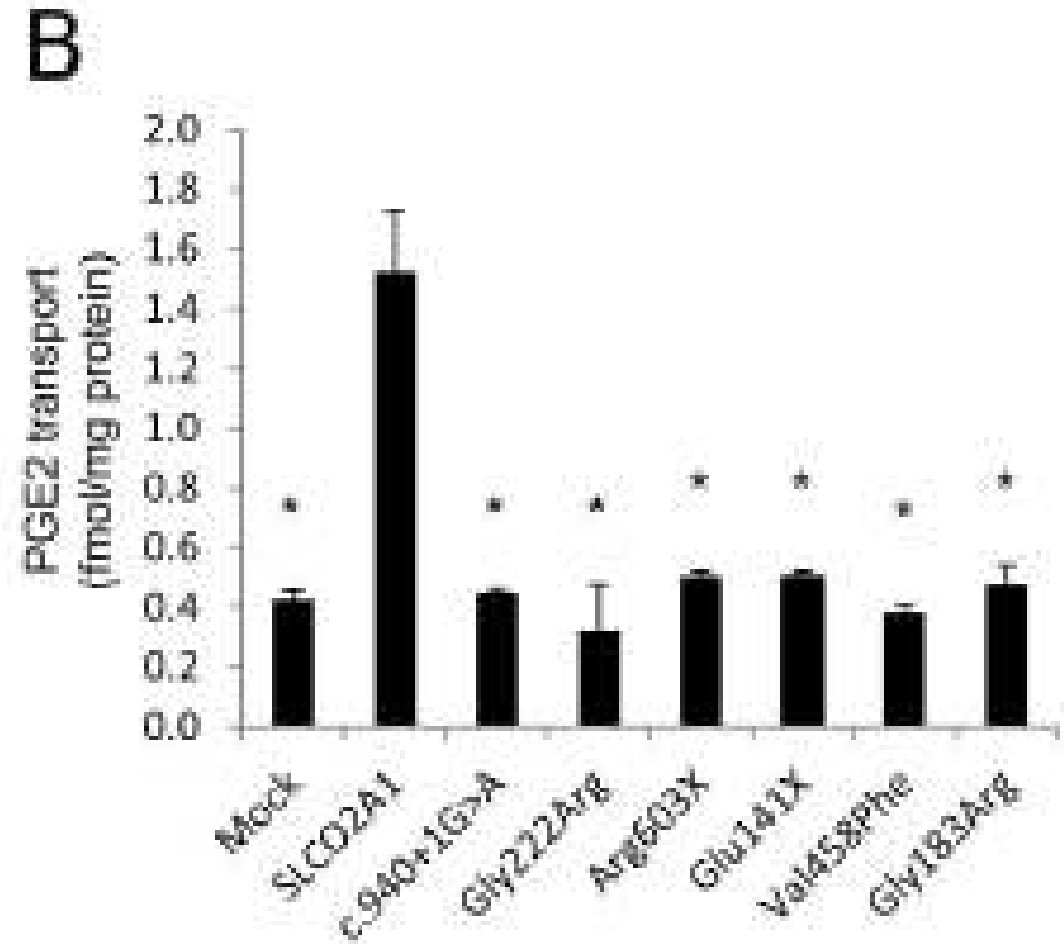
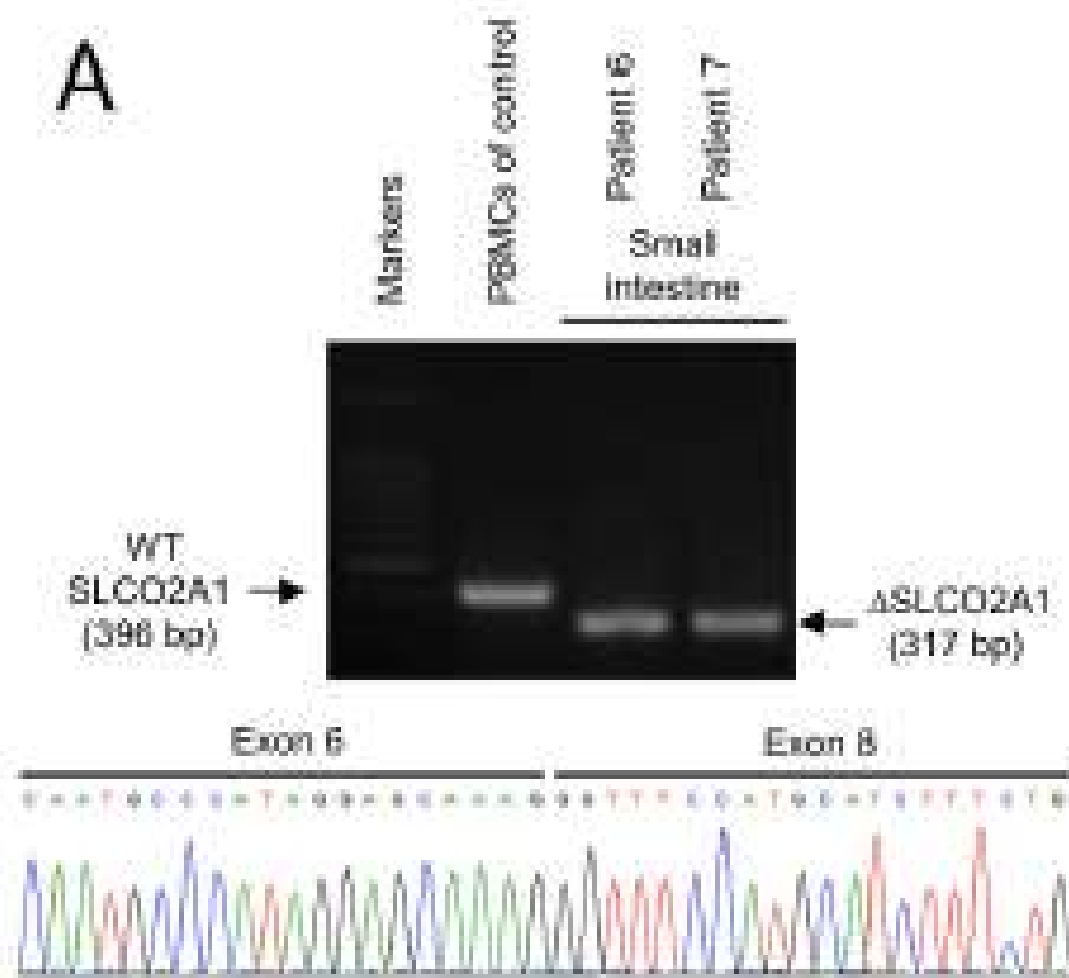
§The 547G>A mutation was identified by Sanger sequencing in a genetic screening of Crohn's disease patients.

CNSU, chronic nonspecific multiple ulcers of the small intestine; HGVD, Human Genetic Variation Database for the Japanese population; CD, Crohn's disease.

doi:10.1371/journal.pgen.1005581.t002

A Hereditary Enteropathy Caused by Mutations in the *SLCO2A1* Gene, Encoding a Prostaglandin Transporter. PLoS Genet. 2015

PGE2トランスポートの機能喪失に関連する



A Hereditary Enteropathy Caused by Mutations in the SLCO2A1 Gene, Encoding a Prostaglandin Transporter. PLoS Genet. 2015

小腸疾患の診断と治療 各論

- ✓血管性病変
- ✓腫瘍性病変
- ✓炎症性疾患（粘膜病変）
- ✓その他

Small Intestinal Bacterial Overgrowth(SIBO)

小腸內細菌增殖症

Diagnostic test	Substrate and characteristic	Testing protocol	Test interpretation ^a	Diagnostic performance	Limitations
Small bowel aspirates/ culture					
Duodenal ^{6,8}	Upper endoscopy performed to obtain samples from the third or fourth portion of the duodenum	3-5 mL of duodenal liquid aspirate using an aseptic technique (e.g., sterile gloves) Samples sent to the laboratory immediately after collection	Positive test: $\geq 10^3$ cfu/mL 65.5% agreement between GBT and duodenal aspirate/culture	Currently, the most widely accepted best diagnostic method for SIBO	Testing is invasive, time intensive, and expensive
Jejunal ⁸	Upper endoscopy performed to obtain samples from the proximal jejunum	≥ 2 mL of jejunal liquid aspirated Samples sent to the laboratory in sterile fashion immediately	Positive test: $\geq 10^3$ cfu/mL	Currently, the most widely accepted best diagnostic method for SIBO	Testing is invasive, time intensive, and expensive

^aGlucose dose of 50 g with rise in hydrogen >12 ppm from baseline is also considered positive.

^bNorth American consensus no longer considers 2 peaks on a lactulose breath test a requirement for establishing a diagnosis of SIBO (8).

The table created with data from Rezaie A, Buresi M, Lembo A, et al. Hydrogen and methane-based breath testing in gastrointestinal disorders: The North American Consensus. *Am J Gastroenterol* 2017;112:775–84 (8); Erdogan A, Rao SS, Gulley D, et al. Small intestinal bacterial overgrowth: Duodenal aspiration vs glucose breath test. *Neurogastroenterol Motil* 2015;27:481–9 (6); and Romagnuolo J, Schiller D, Bailey RJ. Using breath tests wisely in a gastroenterology practice: An evidence-based review of indications and pitfalls in interpretation. *Am J Gastroenterol* 2002;97:1113–26 (46).

GBT, glucose breath test; SIBO, small intestinal bacterial overgrowth.

SIBO 検査方法



Figure 1. Description of the procedure for duodenal aspiration, specimen collection, and handling: The technician flushes the scope with sterile water and prepares a sterile field. A Liguory catheter with a stopcock is assembled (**a**). The scope is passed into the second/third portion of the duodenum with minimal air insufflation and suctioning. The endoscopist and the technician wear sterile gloves and advance the Liguory catheter through the biopsy channel. The technician performs gravity-assisted aspiration by holding the syringe at a height lower than the patient to aid fluid flow. Using gentle suction, ~3 mL of duodenal fluid is collected and immediately transferred to the microbiology laboratory (**b**).

**Small Intestinal Bacterial
Overgrowth: Clinical
Features and Therapeutic
Management Clinical and
Translational
Gastroenterology 2019**

SIBO 判定方法

Small Intestinal Bacterial Overgrowth: Clinical Features and Therapeutic Management Clinical and Translational Gastroenterology 2019

Diagnostic test	Substrate and characteristic	Testing protocol	Test interpretation ^a	Diagnostic pe
Breath test				
Glucose ^{B,46}	Monosaccharide Absorbed in the proximal small intestine	75 g of glucose in 250 mL of water Breath samples are collected at baseline and every 15 min for 90–120 min and measured for hydrogen and methane	Rise in hydrogen ≥ 20 ppm from baseline Or Rise in methane ≥ 10 ppm from baseline Or Rise in hydrogen and methane combined ≥ 15 ppm from baseline	Sensitivity: 20%–93 Specificity: 30%–86
Lactulose ^{B,46}	Disaccharide - Nonabsorbable - Reaches the colon - Used to measure orocecal transit in some cases	10 g of lactulose Breath samples are collected at baseline and every 15 min for 180–240 min for hydrogen	Interpretation of the test results requires reliable differentiation of the colonic peak from the small intestine peak ^b Positive test: increase from baseline ≥ 20 ppm H ₂ by 90 min	Sensitivity: 31%–68 Specificity: 44%–10
Fructose ^B	Monosaccharide - Absorbed in the proximal small intestine - Suitable for patients with diabetes	25 g of fructose in 250 mL of water Breath samples are collected at baseline and every 15 min for 180 min and measured for hydrogen and methane	Rise in hydrogen ≥ 20 ppm above baseline Or Rise in methane ≥ 10 ppm above baseline Or Rise in hydrogen and methane combined ≥ 15 ppm from baseline	Sensitivity: 25%–71 Specificity: 42%–92

^aGlucose dose of 50 g with rise in hydrogen >12 ppm from baseline is also considered positive.

^bNorth American consensus no longer considers 2 peaks on a lactulose breath test a requirement for establishing a diagnosis of SIBO (8).

The table created with data from Rezaie A, Buresi M, Lembo A, et al. Hydrogen and methane-based breath testing in gastrointestinal disorders: The North American Consensus.

Rao SS, Gulley D, et al. Small intestinal bacterial overgrowth: Duodenal aspiration vs glucose breath test. *Neurogastroenterol Motil* 2015;27:481–9 (6); and Romagnuolo.

a gastroenterology practice: An evidence-based review of indications and pitfalls in interpretation. *Am J Gastroenterol* 2002;97:1113–26 (46).

GBT, glucose breath test; SIBO, small intestinal bacterial overgrowth.

本来ならば小腸でのガス産生量はわずかですが、SIBOでは食事に含まれる発酵しやすい糖質が小腸内で発酵されることがガスの発生が多くなります。このガスは主に水素とメタンですが、血流に乗って全身をめぐり、肺にも到達すると呼気中に含まれるようになります。

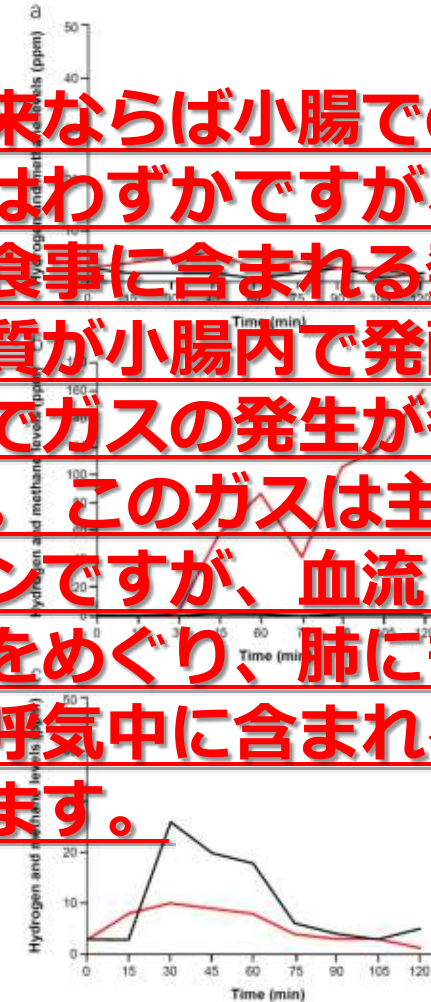
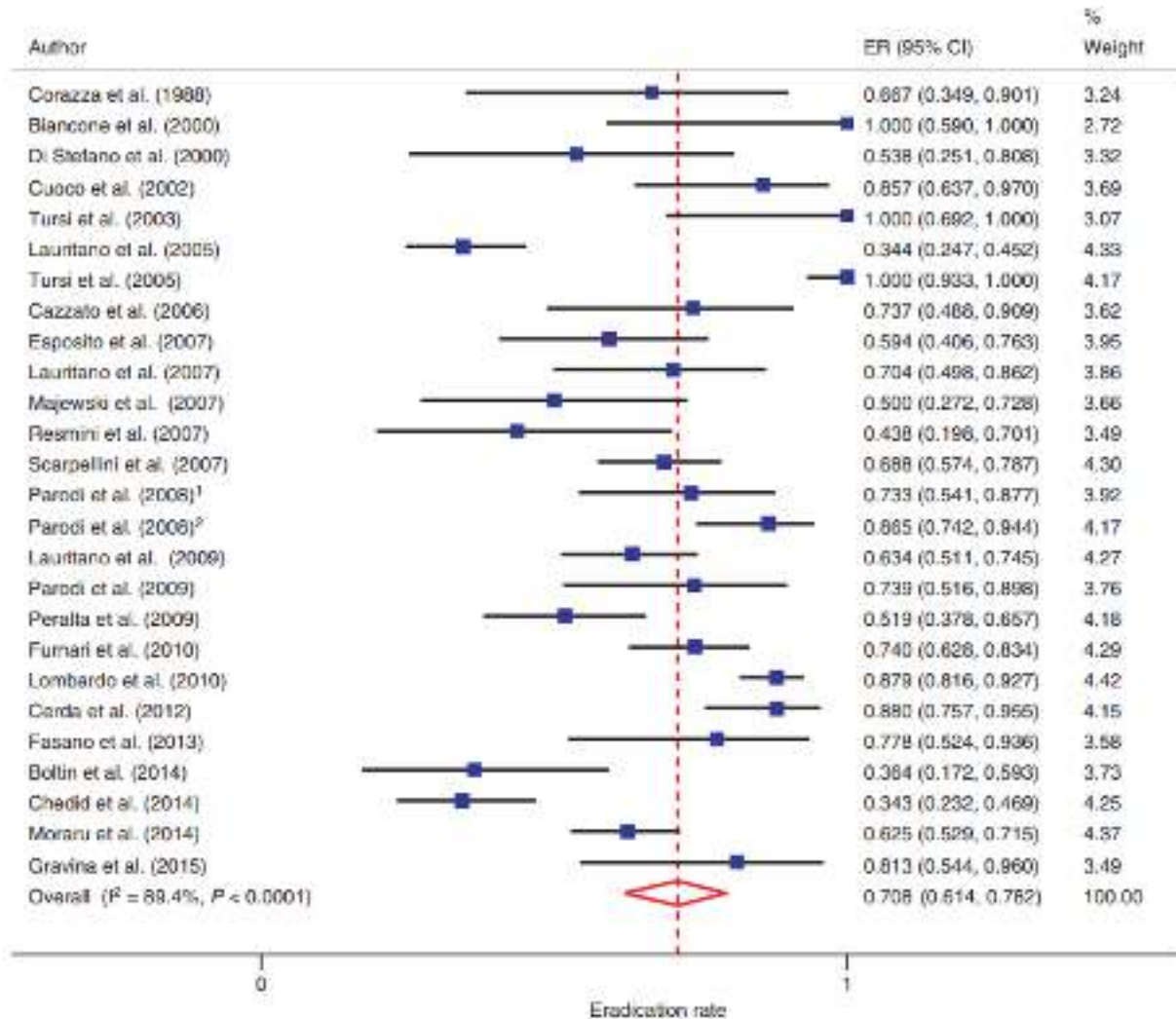


Figure 2. Typical example of breath test results. Shown are a negative breath test result (a), a positive hydrogen breath test showing hydrogen concentration rising >20 ppm from baseline (b), and a positive breath test showing methane concentration rising >10 ppm from baseline (c). Red lines show hydrogen concentrations, and black lines show methane concentrations. Duodenal aspirates and culture results are shown in the text inset of (b).

リファミキシンのSIBOへの有効性



**Systematic review with meta-analysis:
rifaximin is effective and safe for the
treatment of small intestine bacterial
Overgrowth
Aliment Pharmacology and
Therapeutics 2017**

Figure 2 | Forest plot of SIBO eradication rate according to ITT analysis.

本日の内容

- 小腸内視鏡についての最近の知見
- 小腸疾患の診断と治療 各論
- 新しい小腸疾患診療の展開

磁石でコントロールできるカプセル内視鏡で 胃の観察も可能

Table 1. Prevalence of Gastric Focal Lesions Detected by Conventional Gastroscopy and MCE in 350 Patients With Upper Gastrointestinal Complaints, and the Performance of MCE Compared With Gastroscopy

Lesions	Gastroscopy	MCE		
	Patients, n (%)	Patients, n (%)	Sensitivity, % (95% CI)	Specificity, % (95% CI)
Type ^a				
Overall	104 (29.7)	107 (30.6)	90.4 (84.7–96.1)	94.7 (91.9–97.5)
Polyp	43 (12.3)	47 (13.4)	90.7 (82.0–99.4)	96.7 (94.4–98.9)
Ulcer ^b	30 (8.6)	28 (8.0)	90.0 (73.5–97.9)	99.6 (97.6–99.9)
Submucosal tumor	18 (5.1)	17 (4.9)	88.9 (65.3–98.6)	99.6 (97.6–99.9)
Others ^c	13 (3.7)	15 (4.3)	92.3 (64.0–99.8)	98.7 (96.3–99.7)
Location ^d				
Upper stomach	51 (14.5)	54 (15.4)	90.2 (82.0–98.4)	96.7 (94.4–98.9)
Lower stomach	53 (15.1)	53 (15.1)	90.6 (82.7–98.4)	97.9 (96.1–99.7)
Size				
<5 mm	64 (18.3)	71 (20.3)	92.2 (85.6–98.8)	95.1 (92.4–97.8)
≥5 mm	40 (11.4)	36 (10.3)	87.5 (77.3–97.8)	99.6 (97.6–99.9)

^aIf a patient has more than 1 focal lesion, the most important clinical-related finding with the priority of ulcer, submucosal tumor, polyp, and others was chosen as the final diagnosis.

^bIncluding 3 malignant ulcer cases.

^cIncluding early gastric cancer, xanthoma, diverticulum, venous aneurysm, telangiectasia, and ectopic pancreas.

^dUpper stomach includes the cardia, fundus, and body, and lower stomach includes the angulus, antrum, and pylorus.

磁石でコントロールする カプセル内視鏡で胃病変描出

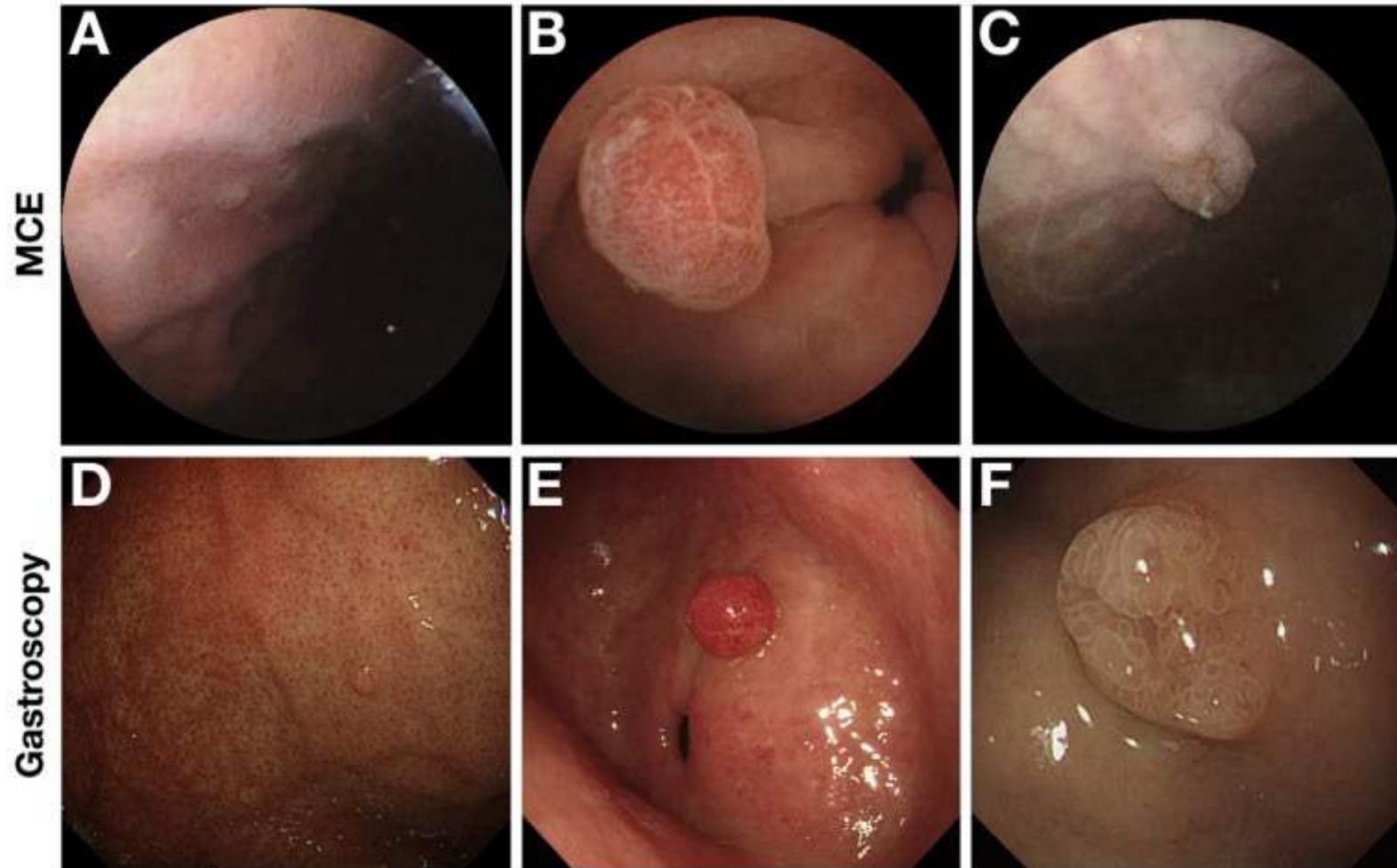


Figure 1. Representative polyps observed on conventional gastroscopy and MCE. (A–C) MCE examination and (D–F) gastroscopy.

磁石でコントロールできるカプセル内視鏡で 胃から小腸の観察が一度で可能

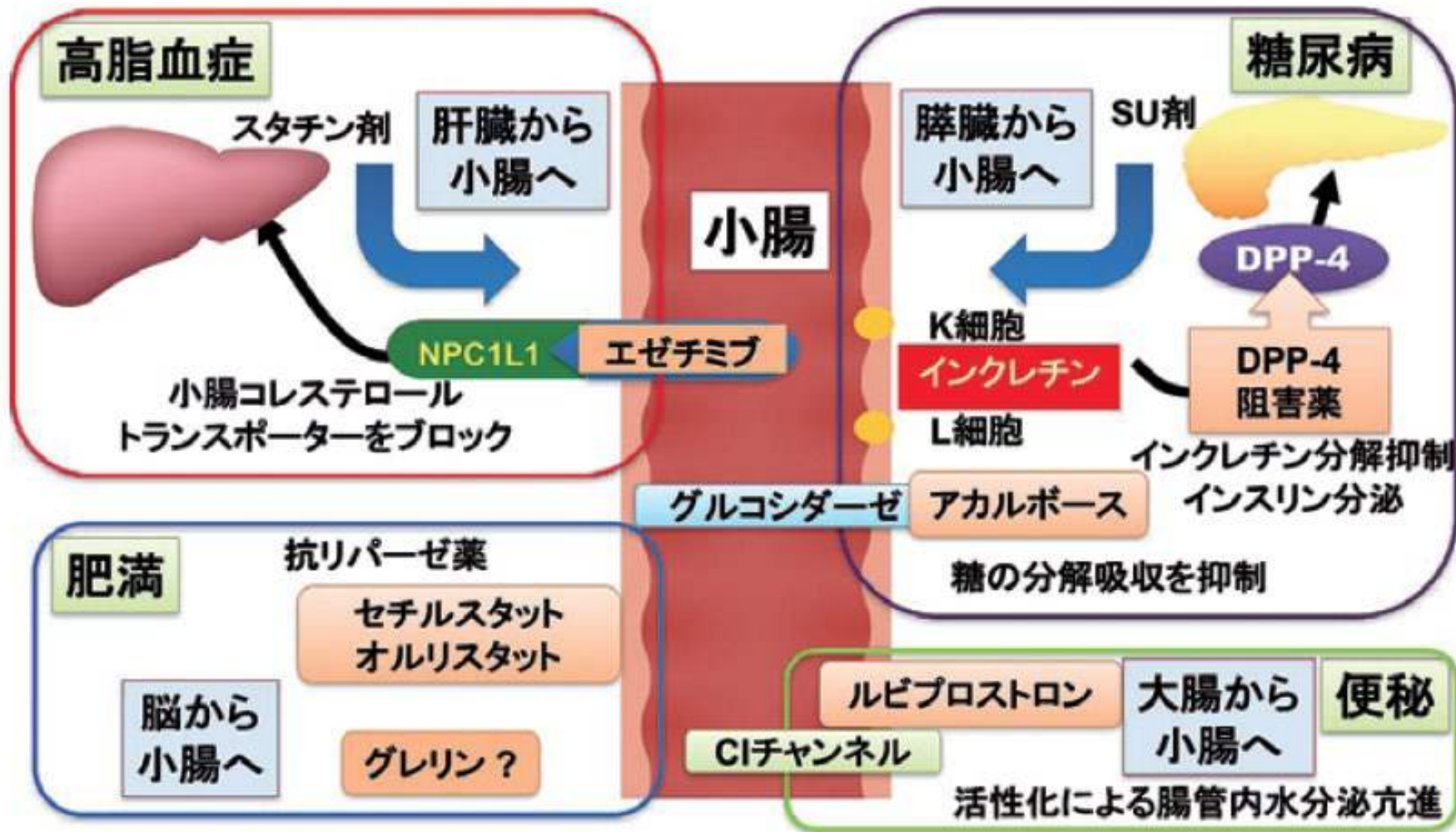
Table 2 Success rate of GSI joint examination

Examination	Success rate, n (%)
Gastric examination	729 (94.92)
Small intestinal examination	748 (97.40)
GSI joint examination	711 (92.58)
GSI, gastro-small intestine	



Fig. 1 Positive lesions detected by MCE. a) Gastric polyp, b) Gastric ulcer, c) Gastric cancer, d) Gastric cancer, e) Small intestine ulcer, f) Small intestine ulcer, g) Small intestine ulcer, h) Small intestine ulcer, i) Small intestine ulcer, j) Small intestine ulcer, k) Small intestine ulcer, l) Small intestine ulcer.

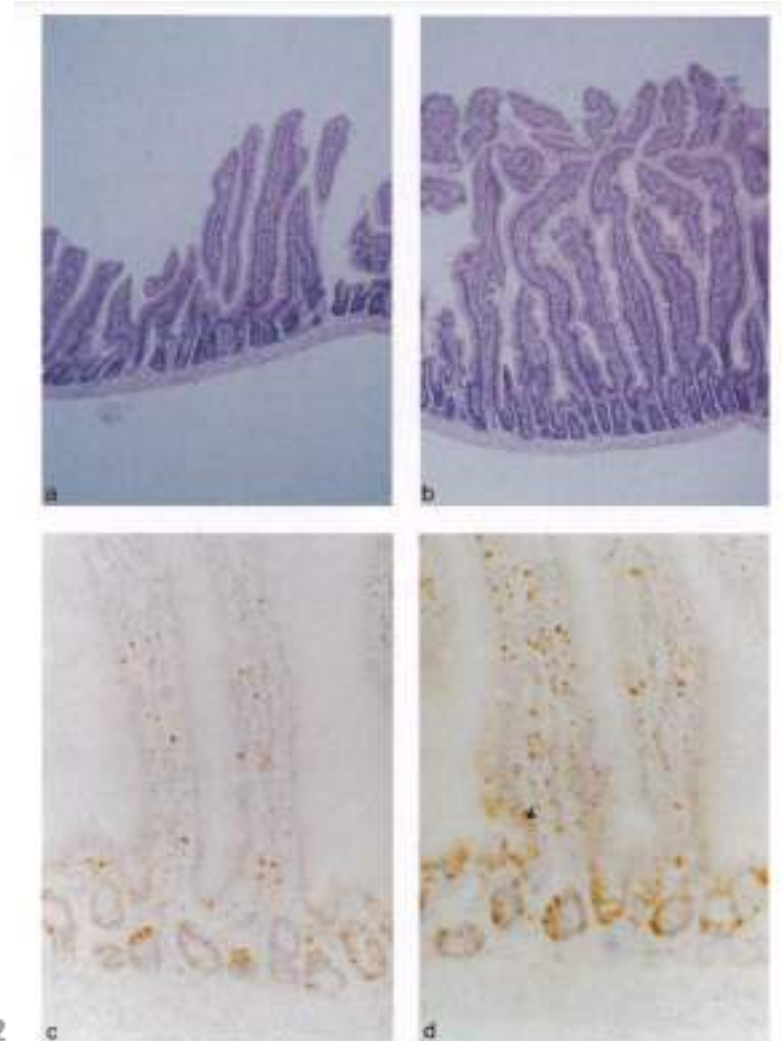
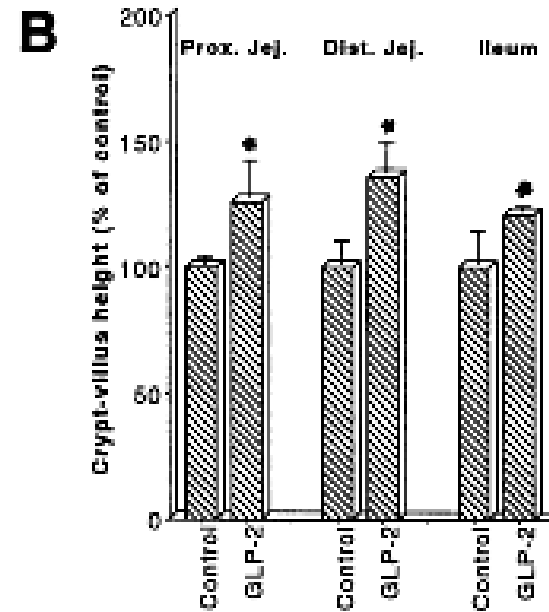
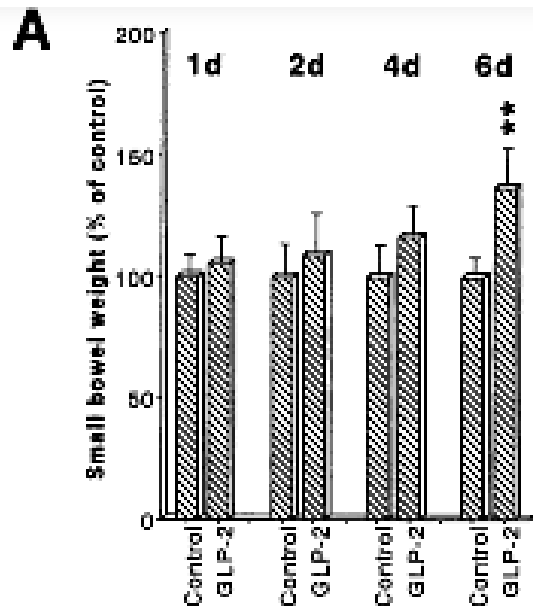
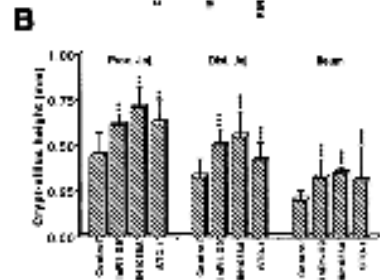
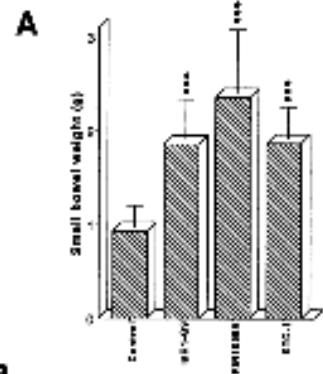
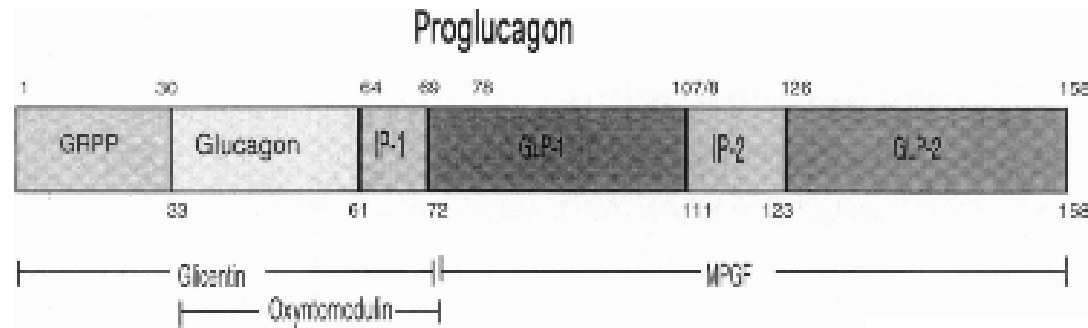
小腸を標的とした全身疾患の治療薬



小腸疾患 (non-CD) を見直す
—小腸病変と全身疾患を見直す—
消化器病学会雑誌 2018

Figure 4. 小腸を標的とした全身疾患の治療薬. 文献 25) より転載.

GLP-2が小腸粘膜過形成を起こす



Induction of intestinal epithelial proliferation by glucagon-like peptide 2
Proc. Natl. Acad. Sci. 1996

Glucagon-like peptide-2 (GLP-2) と小腸の消化・吸収

日本消化器病学会雑誌 2011

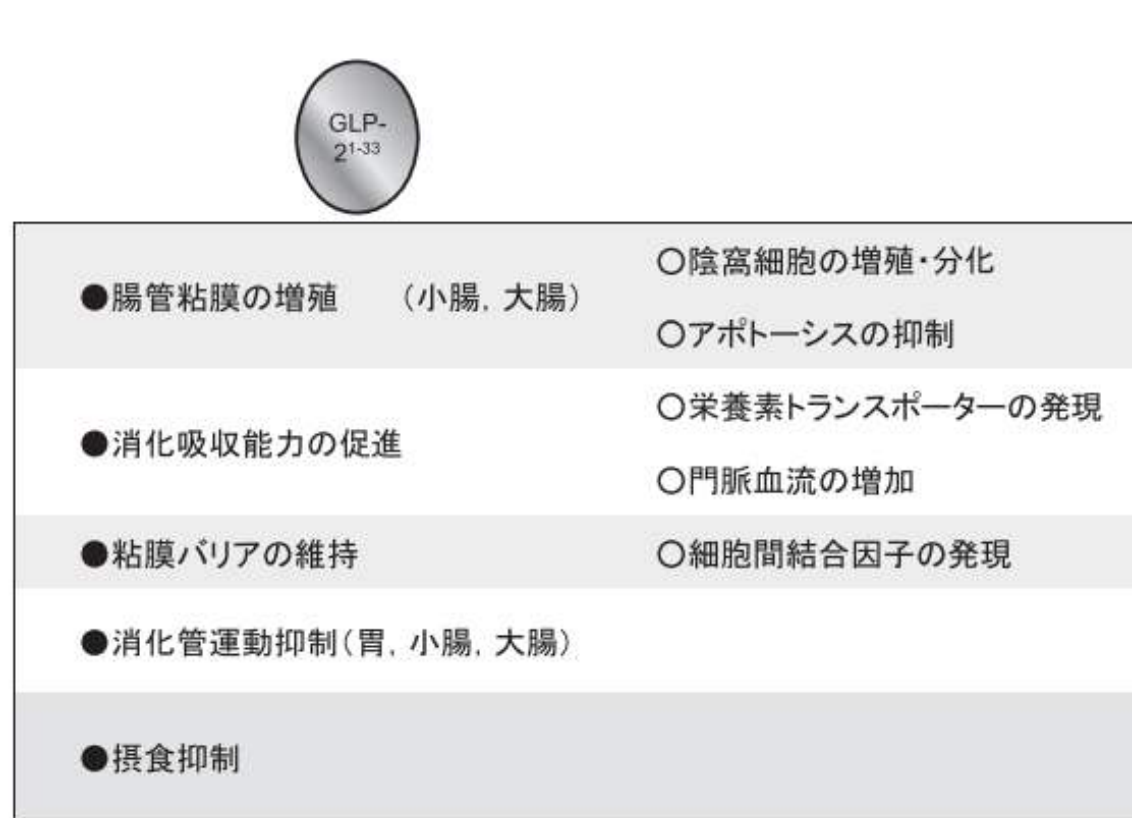
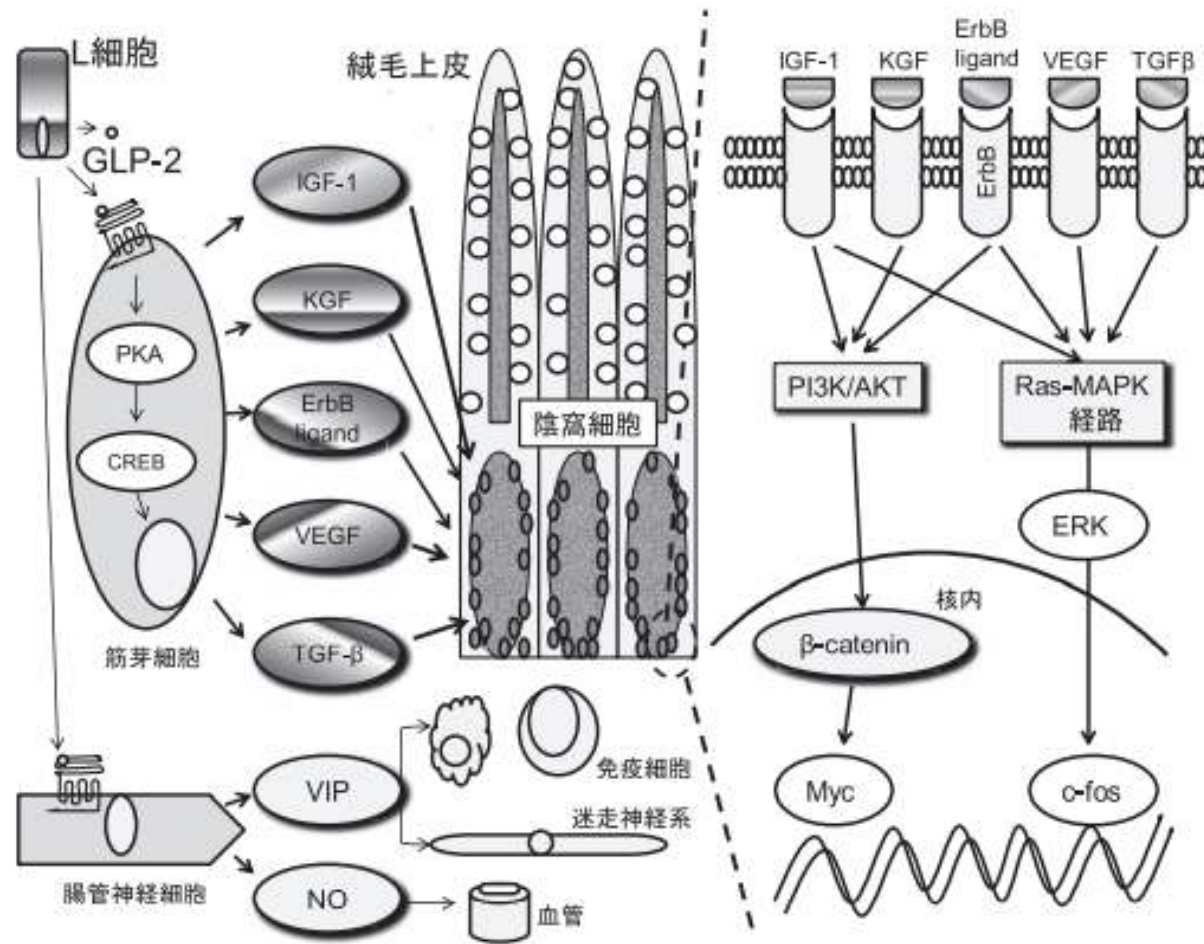


Figure 1. GLP-2 の主な機能.



短腸症候群 GLP-2製剤 栄養サポート率低下効果あり

Table 3 Summary of results for the graded response score (GRS)

	Criterion value	1	2	3	4	5
	0 <20% reduction in parenteral support					Off parenteral support
Placebo	15 (94%)	0 (0%)	1 (6%)	0 (0%)	0 (0%)	0 (0%)
Teduglutide 0.10 mg/kg/day	24 (75%)	2 (6%)	4 (13%)	0 (0%)	2 (6%)	0 (0%)*
Teduglutide 0.05 mg/kg/day	19 (54%)	6 (17%)	6 (17%)	0 (0%)	2 (6%)	2 (6%)

Number (%) of patients within the criterion value groups.

Comparison of 0.10 mg/kg/day of teduglutide vs placebo, $p=0.16$. Comparison of 0.05 mg/kg/day teduglutide vs placebo, $p=0.007$. p Value based on a rank ANCOVA adjusting for multiple comparisons in the primary efficacy analysis.

*One patient was weaned off parenteral support at week 24 with a score of 4.

For GRS score, refer to table 1.

Randomized placebo-controlled trial of teduglutide in reducing parenteral nutrition and/or intravenous fluid requirements in patients with short bowel syndrome Gut 2011

GLP-2製剤 プラセボと比較し 重篤な副作用に差はなし

Table 7 Treatment-emerging adverse events (AEs) and serious adverse events (SAEs)

	Placebo (n = 16)	Teduglutide 0.10 mg/kg/day (n = 32)	Teduglutide 0.05 mg/kg/day (n = 35)
Subjects with AE, n (%)	15 (94%)	31 (97%)	33 (94%)
Subjects with SAE, n(%)	5 (31%)	11 (34%)	13 (37%)
Subjects with any AE or SAE leading to study discontinuation, n (%)	1* (6%)	2* (6%)	6 (17%)
Event description by system organ class			
Cardiac disorders	0	0	1
Cardiac failure congestive	0	0	1
Gastrointestinal disorders	0	2	8
Abdominal distension	0	0	1
Constipation	0	0	2
Haemorrhoidal haemorrhage	0	0	1*
Nausea	0	0	1
Pancreatitis	0	1	0
Small intestinal obstruction	0	1*	0
Vomiting	0	0	1
General disorders and administration site conditions	0	0	1
Asthenia	0	0	1
Infections and infestations	1*	1*	0
Catheter sepsis	1*	1*	0
Investigations	0	0	1
Drug level increased	0	0	1
Nervous system disorders	0	0	3*
Coma	0	0	1*
Dysgeusia	0	0	1*
Hypersomnia	0	0	1*

If a subject experienced more than one adverse event in a category, the subject was counted only once in that category. Each event was counted.

*Serious adverse event. Coma, dysgeusia and hypersomnia were all found in one patient.

Randomized placebo-controlled trial of teduglutide in reducing parenteral nutrition and/or intravenous fluid requirements in patients with short bowel syndrome Gut 2011